PEARSON EDEXCEL INTERNATIONAL ADVANCED LEVEL

Information Technology · Unit 1: Information Technology Systems

TOPIC 2

NETWORKS

Teacher edition — clear notes, original diagrams, step-by-step exercises and exam practice

Specification references 2.1.1 – 2.3.2 · Pearson IAL IT Student Book, Topic 2 (from page 31)



Components · Transmission media · Network models · Metrics · Communication methods · Protocols & OSI · IP & MAC addressing · Network design · Security

TEACHER EDITION — ALL BLANKS FILLED, ANSWERS AND MARK SCHEMES INCLUDED. DO NOT DISTRIBUTE.

Name _____ Class _____ Teacher _____

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Page numbers update automatically. Spec references are to the Pearson Edexcel International Advanced Level in Information Technology (Unit 1).

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SECTION 0

Getting started

> Connection and communication are a big part of everyday life. Every message you send, video you stream and payment you make travels across a network. In this topic you will learn what networks are made of, how data moves across them, how their performance is measured, which rules (protocols) they follow, how to design one, and how to keep one secure.

LEARNING OBJECTIVES — BY THE END OF THIS TOPIC YOU SHOULD BE ABLE TO

- explain the **role of each network component** (NIC, hub, switch, router, modem, gateway, bridge, wireless access point, repeater, server, multi-function device)
- describe how the different **transmission media** are used and how they work — wired and wireless
- compare the features and functions of different **network models**: client–server, peer-to-peer, ad hoc and tethering
- explain the **metrics** used to measure network performance, and carry out simple calculations with them
- describe the purpose and features of **communication methods**: Ethernet, Wi-Fi, Bluetooth, cellular, infrared and ZigBee
- explain **network standards and protocols** — the OSI model, TCP/IP, VOIP and SIP — and the role of **IP and MAC addresses**
- **design a network** to meet a set of requirements, taking account of where devices are located
- explain **why networks must be kept secure**, the threats they face and the hardware and software solutions used to protect them

How to use this handbook

You will see…	What it means
Blanks _____	Fill these in while your teacher explains the idea — not before. Writing the word at the moment you understand it helps you remember it.
Ruled lines	Space for your own answers. The number of lines is a hint about how much detail is expected.
Squared boxes	Space for diagrams. Network diagrams in the exam must be *labelled* — devices *and* the connection/protocol used.
Key terms	Definitions you must be able to give word-for-word-ish in the exam.
Fact check ✔	Places where the textbook simplifies, is out of date or contains a slip. We tell you what the textbook says, what is actually true, and what to write in the exam.
Exercises	Step-by-step practice. Activities marked `Textbook Activity n` come from the student book.

Quick check	Short recall questions at the end of each section. Try them without looking back.
Exam practice	Questions written in the style and mark tariff of the Unit 1 paper (2 hours, 80 marks).

EXAM TIP — READ THE MARKS

The mark tariff tells you how many separate points the examiner wants. A 4-mark "explain" question needs about four linked points — for example two reasons, each with a development ("...because...", "...so that..."). Writing the same point three ways scores one mark.

Getting started activity Textbook p.32

Work with a partner. Write each key term below on a separate piece of paper, fold them and put them in a container. Take turns pulling one out. You may use **one sentence** to explain it; your partner has **two guesses**.

network · device · local area network (LAN) · internet · Wi-Fi · Bluetooth · router

Write your one-sentence description of the term you found **hardest** to explain:

e.g. Router: a device that reads the IP address in each packet and forwards it along the best route to another network such as the internet.

Which term did your partner describe best, and why was their description good?

Answers will vary — good descriptions are short, accurate and say what the thing DOES (its role), not just what it looks like.

What is a network? Textbook p.32

A **network** is simply **two or more devices connected together** so that they can share data and resources. Two laptops joined by a cable are a network; so is the internet, which joins billions of devices.

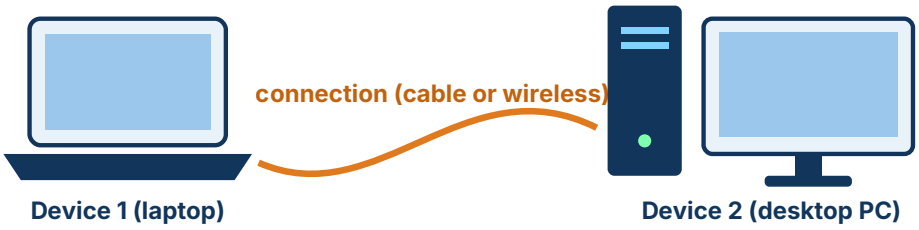


Figure 0.1 The simplest possible network: two devices and one connection

KEY TERMS

LAN	**Local area network** — a network normally located within a single building or site.
WAN	**Wide area network** — a network spread across several buildings, sites or countries. The largest WAN is the internet.
Protocol	A set of rules that govern how data is transferred.

THINK ABOUT IT

List three networks you used today and what each one let you do.

e.g. Home Wi-Fi — streamed music; mobile (4G/5G) network — sent WhatsApp messages; school LAN — logged on and printed homework.

SECTION 1 · SPEC 2.2.3 · TEXTBOOK PP.32–37

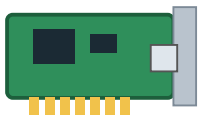
Network components

> A network is built from pieces of hardware, each with a defined job. In the exam you must be able to say *what each component does* (its role), *how it does it* (for example, by reading a MAC address or an IP address) and *why it is used* instead of something else.

KEY TERMS FOR THIS SECTION

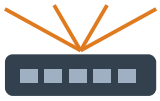
Data packet	The unit of data that carries information across a network. It contains the data itself plus control information such as the source and destination addresses.
Data collision	When two devices on a network send data at the same time and the packets collide, causing errors and re-sends.
MAC address	A unique hardware address given to the NIC in every device at the point of manufacture. Used by switches and bridges.
IP address	A unique address given to a device when it connects to a network. Used by routers.
Routing table	A database stored in a router that holds the known routes across a network to different devices/networks.
Ethernet	A protocol used to connect devices in a LAN using a wired connection.

1.1 The components and their roles



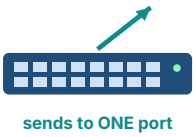
Network interface card (NIC / WNIC) uses MAC address

The component that **every** device needs to connect to a network. It is a circuit board built into the device (or a chip on the motherboard). A NIC normally provides a wired **Ethernet** connection; a **wireless NIC (WNIC)** provides a wireless connection. Each NIC has its own unique **MAC address** set when it is made. p.33



Hub

The most basic way of joining several devices in a **LAN**. A hub receives a data packet and forwards it to **every** device connected to it. Only the intended destination accepts it; the others ignore it. Devices connect by wired Ethernet (or USB). Because every packet goes everywhere, hubs create lots of **unnecessary traffic** and many **data collisions**, which cause errors and reduce performance. p.33



Switch reads MAC address

Developed as an improvement on the hub, a switch also connects devices in a LAN. The big difference: it sends each packet **only to the intended destination**. It records the **MAC address** of every device attached to each port in a table, reads the destination MAC address in the packet and forwards it to that port only. This **reduces traffic**, **reduces collisions** and so **improves performance**. p.34



Router reads IP address

Forwards data packets **between networks** (for example from your home LAN to the internet). It reads the destination **IP address** stored in the packet, looks it up in its **routing table**, and sends the packet along the **most suitable route** towards its destination. p.34



Modem (modulator–demodulator)

Converts analogue signals to digital and digital to analogue. For example, it takes analogue signals sent along telephone wires and converts them into digital signals a computer can process (demodulation), and converts the computer's digital signals back to analogue for transmission (modulation). p.34

✔ FACT CHECK — MODEMS

The textbook says a modem is built into "all modern devices that can be connected to the internet". In practice, a *smartphone* does contain a cellular modem, but a typical laptop or desktop does *not* — it uses its NIC/WNIC to connect to a home router, and the **modem is inside that router box** (DSL, cable or fibre). For the exam, describe the modem's *role* (analogue ↔ digital conversion) and you will be safe.



Gateway

Connects one LAN to another LAN, or a LAN to a **WAN**, when the two networks use **different protocols**. Like two people speaking different languages, the networks cannot communicate directly; the gateway acts as a **translator**, converting data from one protocol to the other. p.35



Bridge reads MAC address

Connects one LAN to another LAN to create **one larger LAN**. A university may have separate LANs in different buildings that are too far apart to work as one; bridges join them. It works much like a switch: it reads the destination **MAC address** and, if that device is on the other connected LAN, forwards the packet across the bridge. p.35



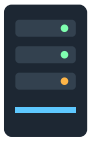
Wireless access point (WAP)

Allows devices with a **WNIC** to connect to a wired network **wirelessly**. It receives the wireless (radio) signals sent by devices and passes the data on to the wired network towards its destination — and the other way round. WAPs are placed so their signal covers the area where users work. p.36



Repeater spec item

Signals get weaker (**attenuate**) the further they travel — along a cable or through the air. A repeater **receives a weak signal and regenerates/boosts it** before sending it on, so data can travel further. It does not read addresses or filter traffic. Example: a Wi-Fi extender upstairs, or a repeater on a long cable run beyond Ethernet's ~100 m limit.



Server spec item

A computer that provides a **service** to other computers (clients) on the network and acts as a point of control. Servers usually have a dedicated role: **file server** (stores and manages files), **web server** (stores web pages and handles requests for them), **print server** (receives print jobs and sends them to the right printer), plus mail, database and authentication servers. See Section 3. p.42



Multi-function device (combined router/switch)

Several components can be combined into one box. The "router" in most homes and small schools is really a **modem + router + gateway + wireless access point**, and usually a **switch** too (the Ethernet ports on the back), often with a basic **firewall**. Every router is set up to act as a gateway — but **not every gateway is a router**. p.37

REMEMBER THE ADDRESSES

Switch and bridge → read the **MAC** address (work *inside* one network). Router → reads the **IP** address (works *between* networks). Hub and repeater → read **no** address at all.

1.2 Hub or switch? Seeing the difference

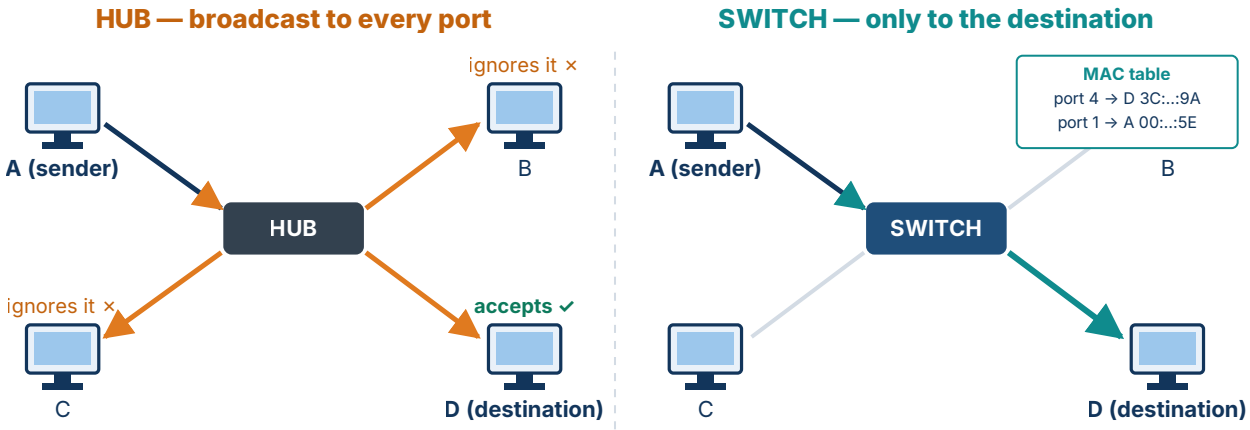


Figure 1.1 A hub floods every port; a switch looks up the destination MAC address and uses one port

1.3 Putting the components together

Figure 1.2 is a typical small organisation network. Trace a packet from a client PC to a website: *client → switch → router → firewall → internet*. Wireless devices join through the Wi-Fi router (which contains a WAP). Textbook Fig 2.1, p.37

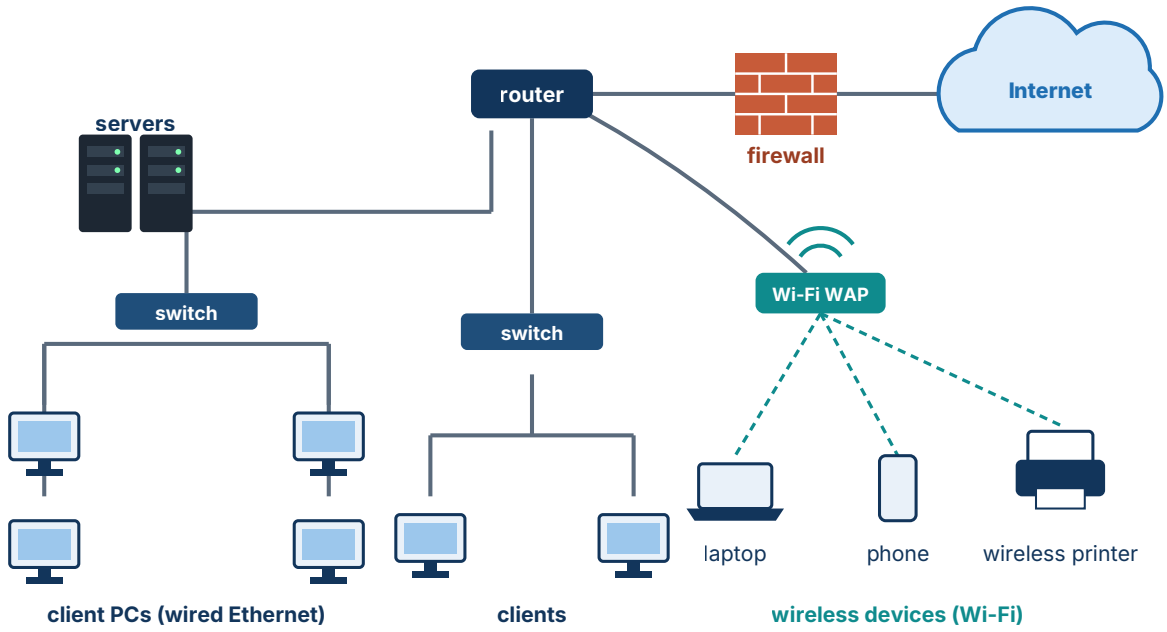


Figure 1.2 An example organisation network (original diagram based on the structure of textbook Figure 2.1)

EXAM TIP

When asked to "describe the role" of a component, give **what it connects**, **what it reads/does to the data**, and **the effect**. For example: "A switch connects devices in a LAN (1). It reads the destination MAC address of each packet (1) and forwards it only to that device (1), reducing traffic and collisions (1)."

EXERCISE 1.1 — Fill in the blanks

Recall

1. Every device needs a **network interface card (NIC)** to connect to a network. A wireless version is called a **WNIC**.
2. A hub forwards each data packet to **every device / all ports** connected to it, which causes unnecessary traffic and **data collisions**.
3. A switch keeps a table of **MAC addresses** and sends each packet only to the **intended destination**.
4. A router reads the destination **IP address** and uses a **routing table** to choose the most suitable route.
5. A modem converts **analogue** signals to **digital** signals and vice versa.
6. A **gateway** is used when two connected networks use different **protocols**; it acts as a translator.
7. A **bridge** joins two LANs to make one larger LAN.
8. A **repeater** regenerates a weak signal so that data can travel further.
9. A **wireless access point (WAP)** lets devices with a WNIC join a wired network wirelessly.

EXERCISE 1.2 — Which component?

Apply

Name the most suitable component for each situation and give a reason.

Situation	Component	Reason
A school's ICT room has 30 PCs that keep slowing down because packets are sent to every PC.	switch	sends packets only to the destination MAC address, so less traffic and fewer collisions
A home network must send data out to the internet and choose the best route.	router	reads destination IP address and forwards between networks using a routing table
Two university buildings each have a LAN; they should work as one LAN.	bridge	joins two LANs into one larger LAN, forwarding by MAC address
Upstairs bedrooms get a very weak Wi-Fi signal.	repeater / Wi-Fi extender	boosts/regenerates the weak signal so it travels further
A company network uses a different protocol from the network of its partner company.	gateway	translates data from one protocol to the other
Staff need to store work centrally so it is backed up every night.	(file) server	stores and manages files centrally; backups made from one place

EXERCISE 1.3 — Compare a hub and a switch

Explain · 4 marks

Explain **two** reasons why a network manager would replace hubs with switches.

1. A hub sends every packet to all connected devices, whereas a switch reads the destination MAC address and sends the packet only to that device (1) — so there is much less unnecessary network traffic (1).

2. Less traffic means fewer data collisions (1), so fewer errors/re-sends and better overall network performance/speed (1).

EXERCISE 1.4 — Inside the home "router"

Textbook Activity 1 · Create

Label the four (or five) components that are usually combined inside a home "router" box, and write one line about each role. Then choose **three** components that interest you and plan a classroom poster in the box below explaining their roles with a diagram.

Component inside the box	Its role in a home network
modem	converts the analogue/broadband line signal to digital and back
router	forwards packets between the home LAN and the internet using IP addresses

switch	the Ethernet ports — connects wired devices, forwarding by MAC address
wireless access point	lets devices with a WNIC connect by Wi-Fi
gateway (and firewall)	links home network to ISP network/protocols; firewall filters traffic

Poster plan (sketch):

QUICK CHECK — Section 1

1. State the role of a NIC.

   Hardware built into a device that allows it to connect to a network (wired or wireless); it holds the MAC address.

2. Which address does a switch use, and which does a router use?

   Switch — MAC address; router — IP address.

3. Give one disadvantage of using a hub.

   Sends data to all devices → unnecessary traffic and more collisions, reducing performance.

4. What is the difference between a bridge and a gateway?

   A bridge joins two LANs (using the same protocols) into one larger LAN; a gateway connects networks that use different protocols and translates between them.

5. Why is it said that "all routers are gateways but not all gateways are routers"?

   A router must translate between the home LAN and the ISP/internet so it is set up as a gateway; but a gateway can simply translate protocols without doing any routing.

SECTION 2 · SPEC 2.2.1 · TEXTBOOK PP.37–41

Network transmission media

> The transmission medium is *what the data physically travels through* — copper, glass, the electrical wiring in the walls, or the air. Each medium has its own speed, range, cost, security and reliability. Choosing the right one is a key part of network design (Section 8).

WIRED MEDIA

- twisted pair cable (copper)
- coaxial cable (copper)
- fibre optic cable (glass/plastic)
- powerline (building wiring)

WIRELESS MEDIA

- microwave
- radio wave
- light (infrared, visible light)
- satellite

2.1 Wired media

Twisted pair cable

The most common LAN cable (the familiar "Ethernet cable" with an RJ45 plug). Inside are **four pairs of copper wires**, each pair **twisted** around each other. Twisting helps cancel out **electromagnetic interference (EMI)** from nearby cables and equipment, and crosstalk between the pairs.

- **UTP** (unshielded twisted pair) — cheap, flexible, most common.
- **STP** (shielded twisted pair) — has an extra foil/braided shield to reduce interference further, e.g. in factories.
- Transmits data as **electrical signals**.
- Maximum recommended run is about **100 m** before the signal weakens.
- Typical speeds: Cat5e up to 1 Gbps; Cat6/Cat6a up to 10 Gbps.

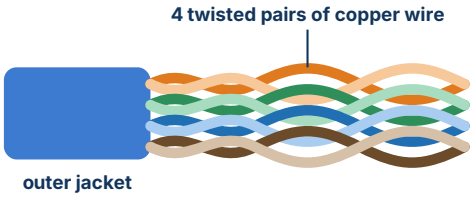


Figure 2.1 Twisted pair cable

Coaxial cable

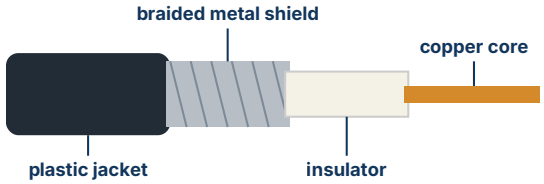


Figure 2.2 Layers of a coaxial cable

A single **copper core** surrounded by an **insulating layer**, a **braided metal shield** and an outer jacket. The shield protects the signal from interference, so coax can carry signals further than UTP with less interference.

- Transmits **electrical signals**.
- Used for **cable TV**, cable broadband and connecting satellite dishes and aerials.
- Thicker and less flexible than twisted pair; now rarely used inside LANs.

Fibre optic cable p.39

A fibre optic cable has many very thin strands of **glass or plastic** that make up the **core**. The core is covered by **cladding**, then a plastic **coating**, then **strengthening fibres** and an outer **cable jacket**. Data is sent as **pulses of light**. The cladding reflects the light back into the core (total internal reflection), so the light bounces along the fibre even round gentle bends.

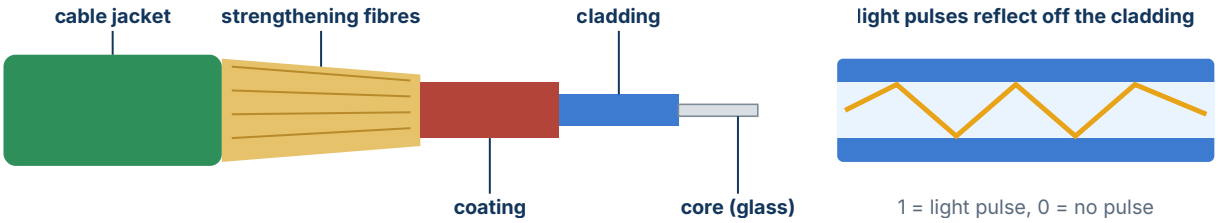


Figure 2.3 Layers of a fibre optic cable (left) and how light travels along the core (right)

Characteristics (textbook)	Why it matters
Transmit data using **light**	Light pulses lose very little strength, so fibre works over many kilometres without repeaters.
Very high speeds see fact check	Ideal for backbones, between buildings, and fibre broadband.
Expensive to manufacture	Higher cost than copper — and needs specialist installers.
Not prone to electromagnetic interference	Light is not affected by electrical noise, so it can run next to power cables and machinery. It is also much harder to tap secretly than copper.
Difficult to install — can be fragile	Glass strands can crack if bent too sharply; joints need special equipment.
Used for internet connections and long distances	Undersea cables that carry most international internet traffic are fibre.

✔ FACT CHECK — FIBRE SPEED

The textbook says fibre can transmit "up to speeds of 1 Gbps". That is a typical *home broadband package*, not the limit of the medium. Data-centre fibre links commonly run at 10–400 Gbps, and a single fibre can carry far more. **In the exam**: say fibre offers the *highest speed/bandwidth of the wired media* over the *longest distances*. If a question quotes a figure, use the figure given.

Powerline p.39–40

Powerline creates a connection using **adapters plugged into ordinary electrical sockets**. Data is sent over the **electrical wiring** already in the building. You need **two adapters**: one beside the router (connected by an Ethernet cable) and one in the room that needs a connection.

- Transmit data using electrical wiring.
- Speeds of up to about 2 Gbps (theoretical; real speeds are much lower and depend on the wiring).
- Relatively expensive to buy compared with a cable.
- Both adapters must be on the **same electrical circuit**.
- Useful where Wi-Fi is weak (thick walls, upstairs office) and running a new cable is impractical.
- Can suffer interference from appliances on the same circuit; should plug directly into the wall, not an extension lead.

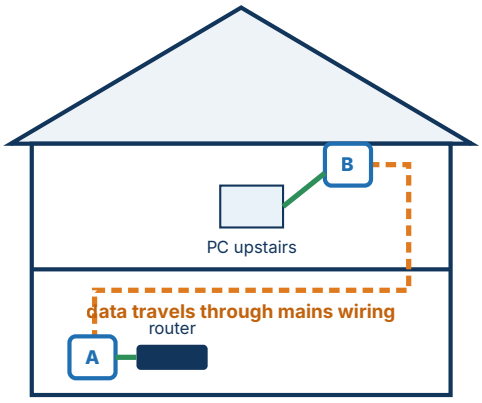


Figure 2.4 A powerline connection between floors

EXAM TIP — DON'T CONFUSE

"Powerline" is a *networking* method using mains wiring. It is **not** the same as Power over Ethernet (PoE), which sends electrical power along an Ethernet cable to devices such as WAPs and IP cameras.

2.2 Wireless media

All the wireless media are forms of **electromagnetic (EM) radiation**. They differ in **frequency** and **wavelength**: the higher the frequency, the shorter the wavelength. Higher frequencies can generally carry more data but travel less well through walls and obstacles.

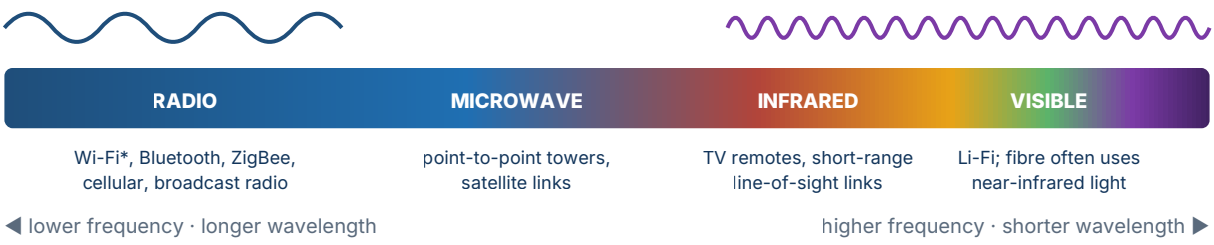


Figure 2.5 Wireless media on the electromagnetic spectrum (not to scale). *Wi-Fi at 2.4/5 GHz is technically in the microwave range; textbooks usually call it "radio waves".

Microwave

Microwaves are high-frequency radio waves (roughly 1 GHz to 300 GHz). For networking they are mostly used for **point-to-point links** between **dishes mounted on towers or tall buildings** — for example to link two office sites, or to connect mobile phone masts back to the network.

- Need a **direct line of sight** between transmitter and receiver — the Earth's curvature limits the distance to roughly 50 km between towers.
- Can carry a lot of data (high bandwidth).
- No cables to lay, so useful across rivers, roads or difficult terrain.
- Signals can be affected by **heavy rain, fog** and obstacles such as new buildings.

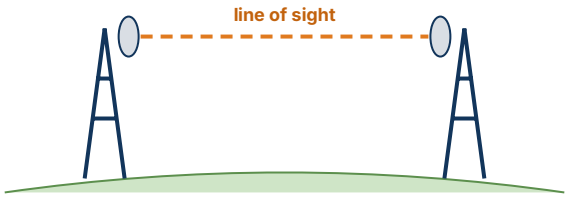


Figure 2.6 A microwave link between two towers

Radio waves

Radio waves have **longer wavelengths** than microwaves. They can travel through walls and other obstacles (although they weaken as they do) and do **not** need line of sight. They are used by most everyday wireless networking: **Wi-Fi, Bluetooth, ZigBee and cellular networks** (Section 5).

- Signals spread out in all directions, so anyone in range can receive them → data should be **encrypted**.
- Range and speed fall as distance and obstacles increase; other devices on the same frequency cause interference.

Light — infrared p.40–41

Infrared has a wavelength **longer than visible light but shorter than radio waves**. It transmits data over **short distances** because it needs a **direct line of sight** and cannot travel through obstacles such as walls.

It has sometimes been used to connect peripherals such as a keyboard and mouse to a computer, but this is **not popular**: the signal is weak and needs line of sight between the peripheral and the computer. Its most common use is the TV remote control. Because infrared cannot pass through walls, signals stay inside the room — a small security advantage.

Visible light can carry data too: *Li-Fi* flickers LED lights faster than the eye can see.

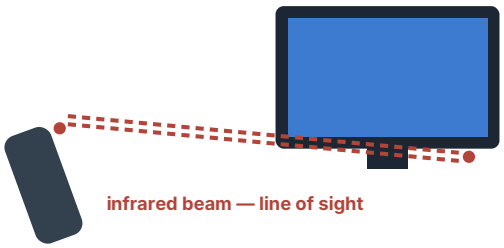


Figure 2.7 Infrared needs a direct line of sight

Satellite p.41

A **satellite** is an artificial body sent into space to orbit the Earth. Satellites transmit data using **radio (microwave) waves**. A base on Earth called a **ground station** sends radio waves up to the satellite (the *uplink*, on the transmitting frequency); the satellite re-transmits them back down to receivers on Earth (the *downlink*, on a retransmitting frequency). The waves pass through every layer of the atmosphere.

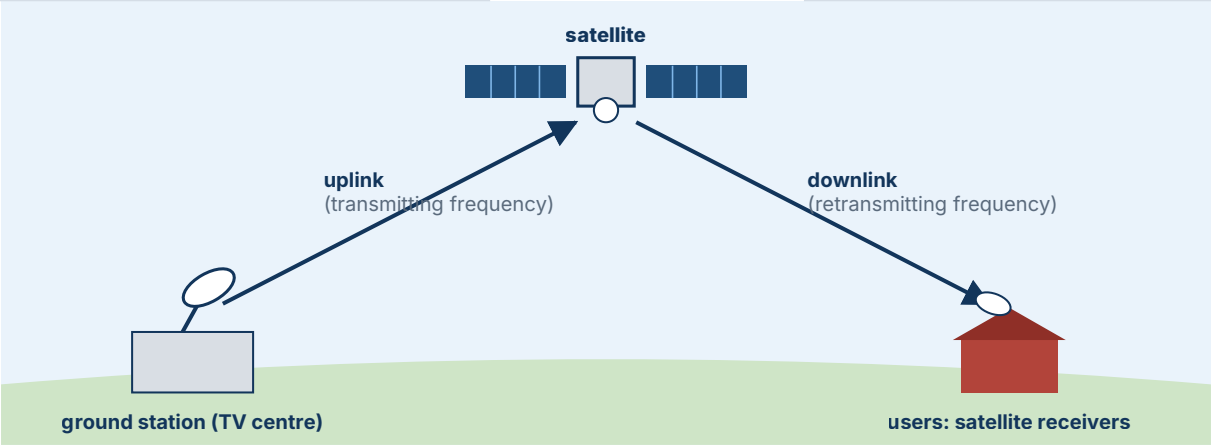


Figure 2.8 Data sent between a ground station, a satellite and users (based on textbook Figure 2.9)

Uses include satellite TV, satellite broadband for remote areas and ships, and **global positioning systems (GPS)**. Satellite links cover huge areas without cables, but signals travel a very long way (a geostationary satellite is about 36 000 km up), so **latency is high** and bad weather can disrupt the signal.

✔ FACT CHECK — GPS

The textbook says GPS uses "triangulation… when at least three satellites are used to pinpoint the location". Two refinements: (1) GPS receivers work out *distance* to each satellite from signal timing, so the method is properly called **trilateration**; (2) a receiver normally needs signals from **at least four** satellites — three for position and a fourth to correct the receiver's clock. Also, a GPS device only *receives*; it does not transmit to the satellites. In the exam, "uses signals from at least three/four satellites to calculate position" will be accepted.

2.3 Comparing transmission media

Medium	Signal	Relative speed	Range	Interference / security	Typical use
Twisted pair	electrical	high (1–10 Gbps)	≈100 m	some EMI; can be tapped	LAN cabling
Coaxial	electrical	medium–high	hundreds of m	shielded; less EMI	cable TV/ broadband
Fibre optic	light	highest	many km	no EMI; hard to tap	backbones, long distance
Powerline	electrical (mains)	medium	same circuit	appliance noise	home, hard-to-cable rooms
Microwave	EM wave	high	≈50 km per hop	line of sight; rain	tower-to-tower links
Radio	EM wave	varies	m to km	through walls; interception	Wi-Fi, Bluetooth, cellular
Infrared	light	low	few m	line of sight; stays in room	remote controls
Satellite	EM wave	medium	global	high latency; weather	GPS, TV, remote internet

EXERCISE 2.1 — Label the cables

Recall

Complete the labels from the outside in.

Fibre optic: cable jacket → strengthening fibres → coating → cladding → core

Coaxial: plastic jacket → braided metal shield → insulator → copper core

Twisted pair cables twist the wires together to reduce electromagnetic interference / crosstalk. Fibre optic cables send data as pulses of light, so they are not affected by electromagnetic interference.

EXERCISE 2.2 — Choose the medium

Apply · justify

Scenario	Best medium	Justification
Linking two hospital buildings 2 km apart; very high data volumes (X-ray images).	fibre optic	works over long distances at very high bandwidth; no EMI from medical equipment
A student's bedroom PC two floors above the router; thick stone walls block Wi-Fi; landlord forbids drilling.	powerline	uses existing mains wiring; no new cables; not blocked by walls
Connecting an oil platform at sea to the company network.	satellite (or microwave)	no cables possible; satellite covers remote areas
Connecting 25 desktop PCs in a classroom to a switch cheaply.	twisted pair (UTP)	cheap, easy to install, fast enough, within 100 m
Joining two offices on opposite sides of a river, 5 km apart, with clear line of sight.	microwave	no cable across river; line of sight available; high bandwidth

EXERCISE 2.3 — Wired vs wireless

Discuss · 6 marks

A café owner wants to give customers internet access and connect two tills. Discuss whether the café should use wired or wireless connections. Consider cost, convenience, security and performance.

Customers: wireless (Wi-Fi) is the only practical option — customers bring their own phones/laptops, no cables or ports needed, can sit anywhere. However radio signals spread outside the café, so the guest network must use WPA2/WPA3 encryption and be kept separate from the business network.

Tills: wired (twisted pair Ethernet) is better — tills don't move; a cable is more secure (cannot be intercepted from outside), more reliable and not affected by interference or busy Wi-Fi traffic from customers; card payment data is sensitive.

Cost: Ethernet cable is cheap but installation in an existing café may need drilling; Wi-Fi needs one WAP.

Conclusion: a mix — Wi-Fi for customers on a guest network, wired Ethernet for the tills.

EXERCISE 2.4 — Revision resource

Textbook Activity 2 · Create

Create a revision resource for the different types of transmission media: flash cards, an audio podcast script, a multiple-choice quiz or a mind map. Plan a **mind map** below, then swap with a partner.

Something in my partner's resource I hadn't thought of:

Something I included that they hadn't:

QUICK CHECK — Section 2

1. State **two** wired and **two** wireless transmission media.

   Wired: twisted pair, coaxial, fibre optic, powerline. Wireless: microwave, radio, light/infrared, satellite.

2. Give two reasons why fibre optic cable is used for long-distance links.

   Light signals lose little strength so travel many km; very high bandwidth/speed; not affected by EMI.

3. Explain why infrared is rarely used to connect a keyboard to a computer.

   Needs a direct line of sight — any object between keyboard and computer blocks it; the signal is weak;

   Bluetooth/radio works through obstacles instead.

4. What must be true about the two sockets used for a powerline connection?

   They must be on the same electrical circuit.

5. Give one drawback of satellite communication.

   High latency (long distance to satellite); affected by weather; expensive.

SECTION 3 · SPEC 2.1.1 · TEXTBOOK PP.42–46

Network models

> Data can be moved in many ways, but a network also needs a *structure*: who is in control, who stores the data, and who provides the services. There are four main ways a network can be structured: **client–server**, **peer-to-peer**, **ad hoc** and **tethering**.

KEY TERMS

Client	A computer in a client–server network that is *not* a point of control; it accesses data and services on a server.
Server	A computer that is a point of control in the network — it "serves" the client computers that connect to it.
Client–server network	A network that has central servers working as a point of control and client computers that connect to them.
Peer	A single computer in a peer-to-peer network.
Peer-to-peer network	A network with no server as a point of control; all devices are equal.
Ad hoc network	A temporary wireless network in which devices connect *directly* to each other without a router or access point.
Tethering	Sharing a mobile phone's internet (cellular) connection with another device such as a laptop or tablet.

3.1 Client–server network p.42–43

There are two types of device: **clients** and a **server**. Each client sends **requests** to a server for data or services; the server **responds**. Servers have a **dedicated role**, for example:

- **file server** — stores and manages files
- **web server** — stores web pages and processes requests for them
- **print server** — receives print jobs and sends them to the correct printer
- also: email server, database server, authentication (log-on) server

Very large networks can use a client–server structure; the large amount of traffic created by all the clients is managed by the different types of server.

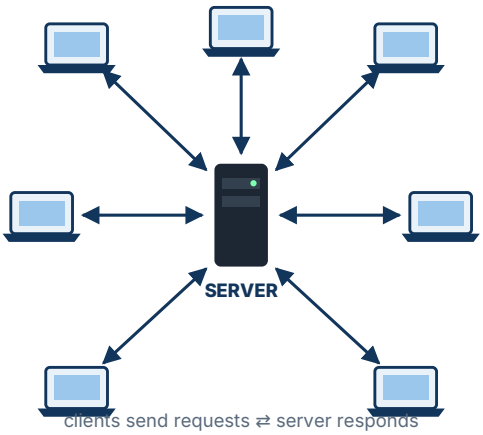


Figure 3.1 A client–server network

TABLE 3.1 BENEFITS AND DRAWBACKS OF A CLIENT–SERVER NETWORK (TEXTBOOK TABLE 2.2)

Benefits	Drawbacks
Security is carried out centrally by the server; each user is not solely responsible for it.	A server is an **expensive** piece of technology; a network may need several, so it is expensive to set up.

Software updates can be carried out from the server, so everyone uses the same version — avoiding compatibility issues.	The network needs **specialist technicians** to maintain the servers, so it is expensive to maintain.
A **backup** of all files can be created centrally from the server; users don't need to remember to back up.	If a **server fails**, clients cannot access the resources and services that server provides (single point of failure).
If a **client crashes**, the rest of the network is unaffected; the user can log on at another client and still reach all their data.	

3.2 Peer-to-peer (P2P) network p.43

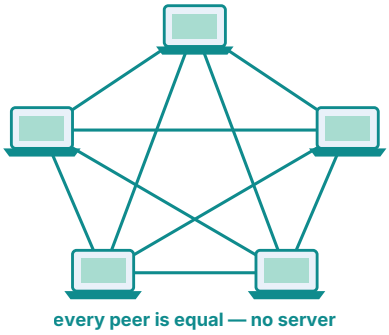


Figure 3.2 A peer-to-peer network

There is only one type of device — a **peer**. The network can be as simple as one phone connected to one computer. There is **no controlling computer at the centre**; every peer has the same level of importance. The main reason for creating a P2P network is to **share data** (and resources such as a printer).

Each user chooses which files and resources to make **public** and which to keep private. Examples: a small home or office network; file-sharing applications; some messaging and blockchain systems.

TABLE 3.2 BENEFITS AND DRAWBACKS OF A PEER-TO-PEER NETWORK (TEXTBOOK TABLE 2.3)

Benefits	Drawbacks
Each peer can **share files and resources** (e.g. a printer) just by setting them to public.	**Less secure** than client–server: each user is responsible for the security of their own computer and data; one careless user can affect others.
Cheap to set up — only transmission media are needed; no expensive servers.	Each user is responsible for **backing up** their own data.
No specialist technicians needed, so **cheap to maintain**.	If too many peers are added, the **volume of traffic can slow** the network.
If one peer fails, only the public files and resources on that computer become unavailable.	

3.3 Ad hoc network

An **ad hoc** network (Latin: "for this purpose") is a **temporary, wireless** network set up for a particular purpose, in which devices connect **directly to one another** — there is **no router or wireless access point** in the middle. It is a wireless form of peer-to-peer networking.

Examples

- Two phones sharing photos directly using Wi-Fi Direct (or a similar direct-transfer feature).
- A laptop connecting straight to a wireless printer or camera to transfer files.
- A group of laptops at a meeting where there is no Wi-Fi.
- Rescue teams creating a network at a disaster site where the infrastructure is destroyed — devices pass messages on for each other (a *mobile ad hoc network*, MANET).

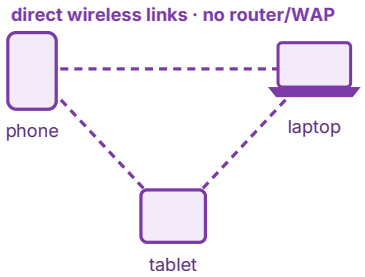


Figure 3.3 An ad hoc network

Benefits of ad hoc networks	Drawbacks of ad hoc networks
Quick and easy to set up anywhere — no infrastructure needed.	Short range — devices must be close to each other.
Cheap: no router, WAP or server to buy.	Less secure: often weak or no authentication; any nearby device may try to connect.
Useful when the normal network is unavailable (travel, emergencies).	Performance drops as more devices join; not suitable for many users.
Devices can join and leave freely.	Temporary: no central management, backup or internet unless one device shares it.

3.4 Tethering p.45–46

Tethering creates a small network of usually **two devices**: a **mobile phone** and another device such as a laptop or tablet. The other device uses the phone's **cellular (3G/4G/5G) internet connection**. The phone can be connected by a **Wi-Fi hotspot**, **Bluetooth** or a **USB cable**.

Users often tether while travelling — e.g. to watch a film on a laptop's bigger screen, or to open emailed documents and edit them on a larger screen when there is no Wi-Fi.

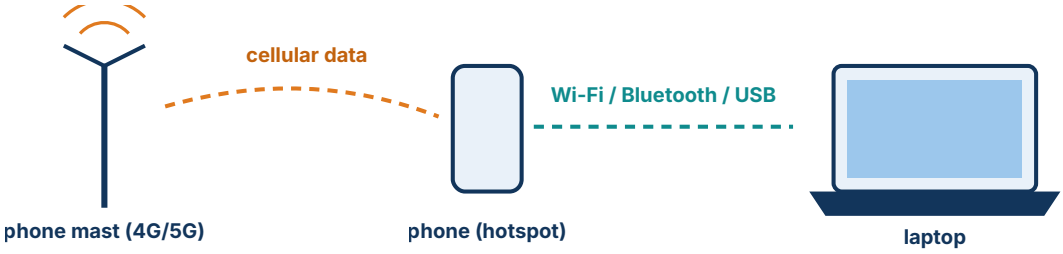


Figure 3.4 A laptop tethered to a mobile phone

TABLE 3.3 BENEFITS AND DRAWBACKS OF TETHERING (TEXTBOOK TABLE 2.5)

Benefits	Drawbacks
Any device can be given an internet connection even when no Wi-Fi network is available.	Drains the **battery**, especially the phone's; without charging, the connection won't last long.
Multiple data plans are not needed — only the phone needs one, saving cost.	Uses the phone's **data allowance**; streaming video or large downloads can use it up quickly.

Can be **more secure than public Wi-Fi**, as the network created is private.	A **strong password** must be set; otherwise someone could join the hotspot and access data.

COMMON CONFUSION — AD HOC VS TETHERING VS P2P

P2P is about *control* (no server; all equal; can be wired or wireless). **Ad hoc** is a *temporary wireless* P2P network with no router/WAP. **Tethering** is about *sharing a phone's cellular internet* with another device. A phone hotspot technically behaves like a mini access point, so it is *not* an ad hoc network.

EXERCISE 3.1 — Which model?

Apply

Situation	Model	One reason
A secondary school with 900 students who must log on to any PC and see their own files.	client–server	central log-on, file storage and backup
Three friends at home sharing one printer and some photos.	peer-to-peer	cheap, no server, easy sharing
A salesperson on a train needs to email a quote from their laptop.	tethering	use phone's 4G/5G; no Wi-Fi available
Two students in a field want to swap a video file between their phones.	ad hoc	direct temporary wireless link; no router
A bank with branches nationwide that must keep customer data secure.	client–server	central security, access rights, backup

EXERCISE 3.2 — Spot the model

Analyse

In the box, draw a **client–server** network with a file server and a print server and four clients. Label each device. Then answer the questions.

1. What happens if the file server fails?

   Clients cannot access the files stored on it (single point of failure); printing via the print server still works.

2. What happens if one client fails?

Nothing else is affected; the user logs on at another client and accesses all their data.

EXERCISE 3.3 — Exam-style

Explain · 2 marks

Hamid is on holiday. There is no Wi-Fi in his hotel, but he needs to use his laptop to upload homework. Explain how Hamid could connect his laptop to the internet.

He could tether the laptop to his mobile phone (1) — turning on the phone's hotspot and connecting the laptop by Wi-Fi/Bluetooth/USB so the laptop uses the phone's 4G/5G cellular data connection (1).

Mark scheme (based on Pearson WIT11 June 2024 Q2(d)): tethering / mobile hotspot (1); uses the cellular/ 3G/4G/5G network to access the internet (1).

EXERCISE 3.4 — Evaluate

Evaluate · 8 marks

A small graphic design firm has six employees. It currently uses a peer-to-peer network but is growing to 30 employees. Evaluate whether it should change to a client–server network.

For client–server: security managed centrally — important as client files are confidential; central backups mean work is not lost if one PC fails, and staff do not have to remember to back up; software updates/ licences managed centrally so all designers use the same version (compatibility of design files); with 30 peers, P2P traffic would slow the network; staff could log on at any PC.

Against: servers are expensive to buy; needs a technician (or contract) to maintain — high ongoing cost for a small firm; a server failure would stop everyone accessing files unless there is a backup server.

Conclusion: as the firm grows to 30 staff handling confidential client work, the security, backup and management benefits outweigh the cost — it should move to client–server (possibly starting with one file server and an outsourced technician).

Levels guidance: L3 (7–8) balanced points for and against, applied to the firm (design files, 30 staff, confidentiality), with a supported conclusion. L2 (4–6) some application, conclusion may be weak. L1 (1–3) generic list of benefits/drawbacks.

QUICK CHECK — Section 3

1. Name the four network models.

   Client–server, peer-to-peer, ad hoc, tethering.

2. State the role of a web server.

   Stores web pages and processes requests for them from clients.

3. Give one benefit and one drawback of a P2P network.

   Benefit: cheap — no server/technician. Drawback: less secure / each user backs up own data / slows with many peers.

4. What makes an ad hoc network different from a normal Wi-Fi network?

   Devices connect directly to each other — no router/WAP; it is temporary.

5. Give two drawbacks of tethering.

   Drains phone battery; uses the phone's data allowance; must be password-protected.

SECTION 4 · SPEC 2.2.2 · TEXTBOOK PP.46–48

Network metrics

> How do we know whether a network is performing well? We measure it. A **network metric** is a measurement of one aspect of performance. Network managers use metrics to spot problems, plan upgrades and check that an internet service provider is delivering what it promised. The spec lists nine metrics: speed, bandwidth, throughput, scalability, latency, error rate, packet loss, availability and jitter.

4.1 Speed, bandwidth and throughput

These three are often confused. Learn them as a set.

Metric	Definition	Unit
Speed	The **rate** at which data is transferred from one device to another.	bits per second (bps, Mbps, Gbps)
Bandwidth	The **maximum** amount of data that *can* be transferred from one device to another in a set amount of time — the capacity of the connection.	bps (e.g. "a 100 Mbps line")
Throughput	The **actual** amount of data that *is* successfully transferred from one device to another in a set amount of time.	bps

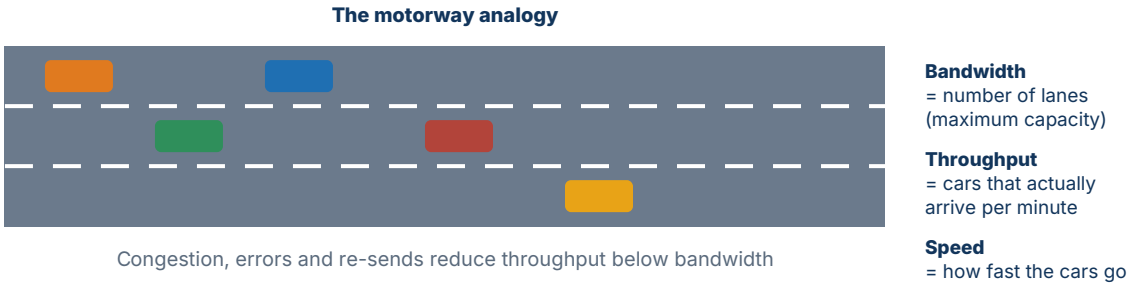


Figure 4.1 Bandwidth is the capacity; throughput is what actually gets through

BITS AND BYTES

Network rates are measured in **bits** per second (lower-case **b**: Mbps). File sizes are measured in **bytes** (upper-case **B**: MB). **1 byte = 8 bits**, so multiply a file size in bytes by 8 before dividing by a rate. Network rates use decimal prefixes: 1 kbps = 1 000 bps, 1 Mbps = 1 000 kbps, 1 Gbps = 1 000 Mbps. In this book, 1 MB = 1 000 KB and 1 GB = 1 000 MB; if an exam question says to use 1 024, follow the question.

WORKED EXAMPLE — DOWNLOAD TIME

A 750 MB video is downloaded on a connection with a throughput of 50 Mbps. How long does it take?
① Convert to bits: 750 MB × 8 = **6 000 Mb** ② Time = size ÷ rate = 6 000 ÷ 50 = **120 seconds (2 minutes)**.
Note: we use *throughput*, not bandwidth — the advertised bandwidth is the best case.

4.2 The other metrics Textbook Table 2.6, p.48

TABLE 4.1 THE MAIN NETWORK METRICS

Metric	Description
Scalability	How well the capacity of a network can be **increased or decreased** to respond to different amounts of data transfer — its ability to handle a larger volume of data than it normally does. One way is increasing bandwidth when necessary. *Example*: an online shop running a sale must scale its network to handle the extra traffic from all the customers.
Latency	The **delay** in the transmission of data across the network. Often measured as the time for data to make a **round trip**: from sender to receiver and back again (e.g. request a web page → page returned). Measured in milliseconds (ms). **Low latency = higher performance.**
Error rate	A measure of the number of errors that occur in a network, usually as a **percentage**. Errors include packet loss, packet corruption and data collisions. Each packet that does not arrive correctly must be **re-sent**, creating more traffic and reducing performance.
Packet loss	A type of error: the **percentage of transmitted packets that were not received** by the receiving device. A packet loss of just 1% can affect performance.
Availability	The amount of time a network is **available for use**, as a percentage of the total time it should have been available. Unavailability may be due to power outages, hardware faults or downtime for updates.
Jitter	The **variation in latency** — how much the delay between packets arriving changes. Caused by congestion, network errors or packets taking different routes. High jitter makes real-time services (VOIP calls, video calls, online games) choppy.

✓ FACT CHECK — LATENCY AND JITTER

Latency: the textbook defines it as round-trip time. Strictly, latency can be one-way; round-trip time (RTT) is what the `ping` command reports. Either is accepted — say which one you mean.
Jitter: the textbook calls it "a measure of the disruption that occurs when data is transmitted". That is vague. Pearson's own mark scheme (WIT11, June 2024, Q1(d)(i)) matches jitter to the **variation in packet arrival timing**. Use that wording in the exam.

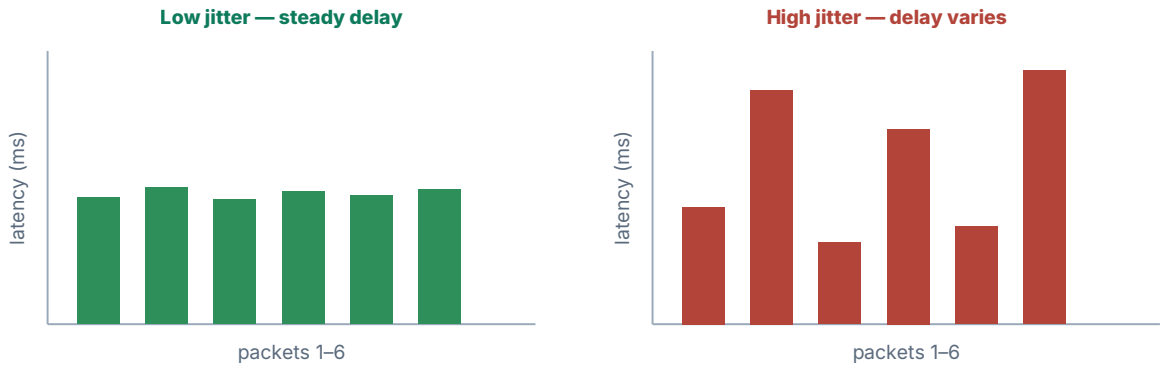


Figure 4.2 Both networks have similar average latency, but the right one has high jitter — a video call would stutter

4.3 Calculating metrics

Metric	Formula
Transfer time	time (s) = file size (bits) ÷ throughput (bits per second)
Packet loss %	(packets sent − packets received) ÷ packets sent × 100
Error rate %	packets with errors ÷ total packets sent × 100
Availability %	time actually available ÷ time it should have been available × 100
Throughput	data successfully transferred (bits) ÷ time taken (s)

EXERCISE 4.1 — Calculations (show your working)

Apply · calculator allowed

1. A 2 GB game update is downloaded with a throughput of 80 Mbps. How many seconds does it take? Give your answer in minutes and seconds.

   2 GB = 2 000 MB × 8 = 16 000 Mb. 16 000 ÷ 80 = 200 s = 3 minutes 20 seconds.

2. A server sent 40 000 packets; 39 400 were received. Calculate the packet loss percentage.

   Lost = 600. 600 ÷ 40 000 × 100 = 1.5%.

3. A school network should be available 12 hours a day for 20 days (240 hours). It was down for 6 hours. Calculate its availability.

   Available 234 h. 234 ÷ 240 × 100 = 97.5%.

4. An ISP promises 99.9% availability over a year (8 760 hours). What is the maximum downtime allowed?

   0.1% of 8 760 = 8.76 hours (about 8 h 46 min).

5. 1.2 GB of data was transferred in 5 minutes on a line with a bandwidth of 100 Mbps. Calculate the throughput and say what percentage of the bandwidth was achieved.

   1.2 GB = 1 200 MB × 8 = 9 600 Mb; 5 min = 300 s. 9 600 ÷ 300 = 32 Mbps. 32 ÷ 100 = 32% of bandwidth.

6. Four ping tests to a server gave round-trip times of 18 ms, 20 ms, 19 ms and 23 ms. Calculate the average latency. Which value suggests jitter?

(18+20+19+23) ÷ 4 = 80 ÷ 4 = 20 ms. The 23 ms result varies most from the others — variation = jitter (range 5 ms).

EXERCISE 4.2 — Match the metric

Recall · 4 marks

Draw a line (or write the letter) to match each description to a metric. This mirrors Pearson WIT11 June 2024 Q1(d)(i).

Description	Metric
The variation in the time packets take to arrive	jitter
The percentage of bits/packets that arrive incorrectly	error rate
The time it takes a packet to travel across the network	latency
The maximum capacity of a channel	bandwidth
The amount of data actually transferred per second	throughput
The ability to grow to cope with more traffic	scalability

EXERCISE 4.3 — Which metric matters most?

Analyse

For each user, state the metric that matters **most** and explain why.

User	Metric	Why
An online gamer playing a fast shooting game	latency	any delay between action and response loses the game
A doctor on a video consultation	jitter (and latency)	varying delay makes video/audio freeze and break up
A video editor uploading 50 GB of footage	throughput / bandwidth	huge file — upload time depends on data per second
An online ticket site on the day concert tickets go on sale	scalability	traffic suddenly rises far above normal
A hospital's patient-record system	availability	staff need records 24/7; downtime risks patient safety

EXERCISE 4.4 — Measure your own network

Investigate

1. On a school or home computer, open a command prompt and type `ping google.com` (Windows) or `ping -c 4 google.com` (macOS/Linux). Record the four times and the packet loss.

Ping 1: ms	Ping 2: ms	Ping 3: ms	Ping 4: ms	Packet loss: %

2. Use a reputable speed-test website. Record download and upload speed. Is this bandwidth or throughput? Explain.

Throughput — it measures the data actually transferred during the test, which is normally below the advertised bandwidth.

3. Repeat at a busy time (e.g. evening). What changed and why?

Usually lower throughput / higher latency in the evening because more users share the network → congestion.

QUICK CHECK — Section 4

1. Explain the difference between bandwidth and throughput.

Bandwidth is the maximum amount of data that can be transferred per second; throughput is the amount actually transferred successfully per second, normally less.

2. Why does a high error rate reduce network performance?

Packets that arrive incorrectly must be re-sent, creating more traffic and congestion, so less useful data gets through.

3. Give two causes of a network being unavailable.

Power cuts; hardware failure; planned maintenance/updates; cyber attack (e.g. DoS).

4. How many bits are in 3 MB?

3 × 1 000 000 × 8 = 24 000 000 bits (24 Mb).

SECTION 5 · SPEC 2.1.2 · TEXTBOOK PP.48–53

Communication methods

> A communication method is a **protocol** — an agreed set of rules — for sending data over a particular medium. Devices can only talk to each other if they use the same protocol. You need to know the **purpose** and **features** of six methods: **Ethernet, Wi-Fi, Bluetooth, cellular, infrared and ZigBee**.

5.1 Ethernet p.49–50

Ethernet is the protocol used to connect devices in a **LAN** using a **wired** connection — usually twisted pair cable, sometimes fibre. It was developed in the 1970s and is standardised as **IEEE 802.3**. Data is sent in units called **frames**, addressed using **MAC addresses**. Speeds have grown from 10 Mbps to 100 Mbps, 1 Gbps and 10 Gbps and beyond.

WHY IS ETHERNET STILL USED? (TEXTBOOK)

- **cheap** to buy (low manufacturing costs)
- **more secure** than wireless — a cable must be physically accessed
- **reliable** — uses error detection to detect errors in transmission
- **compatible** — computers and routers are built with an Ethernet port
- **easy to install** — just plug the cable into the port

Drawbacks: devices are fixed to where the cable runs (no mobility); cabling a building can be disruptive; each run is limited to about 100 m on copper.

5.2 Wi-Fi p.50

Wi-Fi is a protocol (IEEE 802.11) used to transmit data **wirelessly** between devices using **radio waves**. It uses two main frequency bands: **2.4 GHz** and **5 GHz** (newer Wi-Fi 6E and Wi-Fi 7 also use **6 GHz**).

	2.4 GHz	5 GHz
Range	longer (≈45 m indoors)	shorter (≈15 m indoors)
Through walls	better	worse
Speed	lower	higher
Interference	busy — shared with Bluetooth, ZigBee, microwave ovens, neighbours' Wi-Fi	more channels, less crowded

The first Wi-Fi standard (1997) could transfer up to **2 Mbps** using 2.4 GHz. Each new generation has increased speeds. Wi-Fi 6 (802.11ax) has a theoretical maximum of about **9.6 Gbps**, and Wi-Fi 7 (802.11be, certified 2024) is higher still. Real-world throughput is far lower than these theoretical figures.

Wi-Fi devices connect through a **wireless access point** (usually inside the router) and need a **WNIC**. Data should be encrypted with **WPA2** or **WPA3**.

✔ FACT CHECK — WI-FI HISTORY

The textbook says the 5 GHz band "was developed in 2009". In fact the **802.11a** standard used 5 GHz from **1999**. 2009 is when **802.11n** (Wi-Fi 4) was ratified — the first mainstream standard able to use *both* 2.4 GHz and 5 GHz. Exam questions won't ask for these dates; the key point is that 5 GHz is faster but has a shorter range.

5.3 Bluetooth p.50–51

Bluetooth is a protocol for transmitting data wirelessly between two devices that are **close together** (typically within **10 m**). It uses radio waves on the **2.4 GHz** band only. It was invented in **1994** by Ericsson; the first consumer products (hands-free headsets) appeared around 1999–2000.

Devices are connected by **pairing**:

1. On the **host device** (e.g. a smartphone), switch on Bluetooth and scan for devices.
2. A list of compatible devices in range is shown.
3. The user selects the device (e.g. wireless headphones).
4. Sometimes a **PIN code** must be entered or confirmed.
5. The connection is created and remembered for next time.

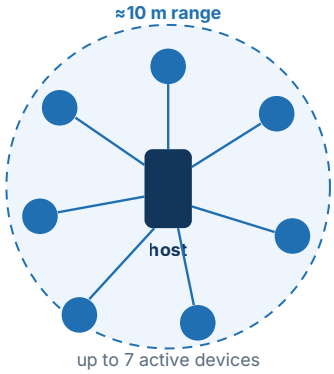


Figure 5.1 A Bluetooth host with paired devices

- Up to **seven** devices can be connected to one host.
- Each must be within about 10 m.
- Transfers data at up to about **3 Mbps**.
- Low power — good for battery devices.

Uses: wireless headphones and speakers, smartwatch ↔ phone, phone ↔ car entertainment system, wireless keyboard and mouse, game controllers.

5.4 ZigBee® p.51

ZigBee is a wireless protocol that uses **radio waves**, normally at **2.4 GHz**, to connect different types of **low-power smart devices**. It is one of the main methods of connection in the **Internet of Things (IoT)**.

- Needs a **ZigBee hub**; all ZigBee devices connect to it so they can communicate.
- Low data rate — up to **250 kbps** (enough for "turn on", "21 °C").
- Range up to about **100 m** (often 10–20 m indoors).
- Very **low power** — sensors can run for years on a battery.
- Devices form a **mesh**: mains-powered devices pass messages on for others, extending the range.

Uses: smart light bulbs, smart plugs, thermostats, door/window sensors, smart meters, touchscreen home-control panels.

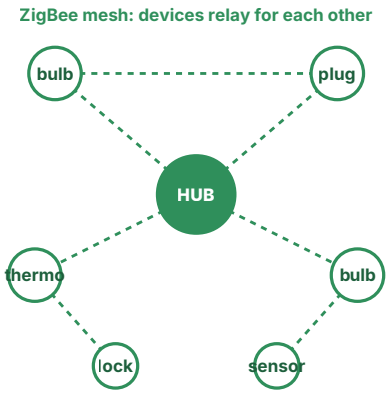


Figure 5.2 A ZigBee smart-home network

✔ FACT CHECK — "IF YOU USED ALEXA, YOU USED ZIGBEE"

The textbook suggests that using a voice assistant to turn on the lights means you have used ZigBee. That is *sometimes* true — some smart speakers contain a built-in ZigBee hub — but many smart bulbs connect by **Wi-Fi** or **Bluetooth** instead, and newer devices may use Thread/Matter. The voice command itself travels over Wi-Fi to the internet. In the exam, link ZigBee to **low-power IoT/smart-home devices using a hub**.

	Bluetooth	ZigBee
Main purpose	connect personal devices (phone ↔ headphones)	connect many low-power smart/IoT devices
Frequency	2.4 GHz	2.4 GHz (also 868/915 MHz in some regions)
Speed	up to ≈3 Mbps	up to 250 kbps
Range	≈10 m	up to ≈100 m; mesh extends it
Structure	host + up to 7 devices (pairing)	hub + many devices (mesh)

5.5 Cellular p.52–53

A **cellular network** lets mobile devices make calls and use data while moving. The area is divided into **cells**, each served by a **base station** (mobile phone mast). Neighbouring cells use different frequencies so they don't interfere. As you move, your phone is **handed over** from one cell's base station to the next. The base stations are linked (by cable, fibre or microwave) to the operator's network and the internet. The original digital standard was **GSM** (2G); later generations are 3G, 4G and 5G.

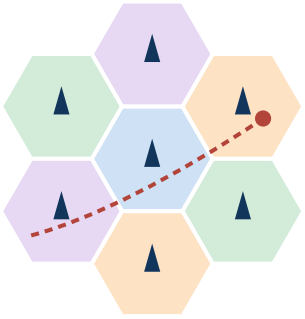


Figure 5.3 Cells and base stations

TABLE 5.1 GENERATIONS OF CELLULAR NETWORK (MAXIMUM SPEEDS AS GIVEN IN THE TEXTBOOK)

Generation	Up to	What it made possible
2G (GSM)	0.1 Mbps	digital calls and SMS text messages; very basic data
3G	42 Mbps	mobile internet browsing, email, apps
4G	150 Mbps	HD video streaming, video calls, VOIP
5G	10 Gbps	very low latency; huge numbers of IoT devices; smart cities

These are the textbook's "up to" figures for common versions of each generation; later versions (e.g. LTE-Advanced) are faster, and everyday speeds are much lower. Use the figures above unless a question gives others.

5.6 Infrared p.53

Infrared communication sends data as pulses of **infrared light** (Section 2.2). It needs a **direct line of sight**, works only over a **few metres**, and cannot pass through walls. The protocol used in many TV-style remote controls is the **NEC infrared transmission protocol** (developed by the Japanese company NEC). An older standard for exchanging data between devices was **IrDA**.

✔ **FACT CHECK**

The textbook writes "national electrical code (NEC) infrared protocol". NEC here is the electronics company **NEC Corporation**, not the "National Electrical Code" (a US wiring safety standard). Just call it the **NEC infrared protocol**.

5.7 Summary of communication methods

Method	Medium	Typical range	Typical max speed	Best for
Ethernet	cable	≈100 m per run	1–10 Gbps+	fixed PCs, servers, printers
Wi-Fi	radio 2.4/5/6 GHz	≈15–45 m indoors	Gbps (theoretical)	laptops, phones, tablets
Bluetooth	radio 2.4 GHz	≈10 m	≈3 Mbps	headphones, watches, peripherals
ZigBee	radio 2.4 GHz	up to ≈100 m (mesh)	250 kbps	smart home / IoT sensors
Cellular	radio (mast)	km per cell	4G 150 Mbps · 5G 10 Gbps	mobile data on the move, tethering
Infrared	infrared light	few m, line of sight	low	remote controls

EXERCISE 5.1 — Fill in the blanks Recall

1. Bluetooth uses the 2.4 GHz frequency and has a range of about 10 m. Up to seven devices can connect to one host.
2. The process of connecting two Bluetooth devices is called pairing.
3. ZigBee devices all connect to a ZigBee hub. ZigBee can transfer data at up to 250 kbps.
4. Wi-Fi at 5 GHz is faster but has a shorter range than Wi-Fi at 2.4 GHz.
5. In a cellular network each area is called a cell and is served by a base station / mast.
6. Infrared needs a direct line of sight between devices.
7. Ethernet is considered more secure than Wi-Fi because a physical cable connection is needed to access it.

EXERCISE 5.2 — Connect the smart home

Apply · 10 marks style

For each connection, state the most suitable communication method and give a reason. In the June 2024 paper, 10 marks were awarded for correctly labelled connections in a network diagram.

Connection	Method	Reason
Desktop PC in a study → router (same room)	Ethernet	fixed device, fast, reliable, secure
Laptop moved around the house → router	Wi-Fi	mobile within the house; range ≈45 m
Smart light bulbs → hub	ZigBee	low power, low data, hub/mesh
Wireless earbuds → phone	Bluetooth	short range personal device, pairing, low power
Phone → internet when out shopping	cellular (4G/5G)	no Wi-Fi; works on the move via masts
Remote control → TV	infrared	short range, line of sight, cheap
Smart thermostat → hub	ZigBee	sends small amounts of data; battery lasts years

EXERCISE 5.3 — Why do calls cut off?

Textbook Activity 7 · Research

Use the internet to find out why mobile phone calls sometimes cut off mid-conversation. What is happening in the cellular network?

Possible reasons: the phone moves out of range of one base station and the handover to the next cell fails; the next cell is congested (no free capacity); the signal is blocked or weakened by buildings, tunnels, lifts, hills; the user is at the edge of coverage between cells; interference; switching between network generations (e.g. 4G to 3G) fails; mast fault/maintenance.

EXERCISE 5.4 — Compare

Explain · 4 marks

Explain **two** reasons why ZigBee rather than Bluetooth is used to connect smart lights throughout a house.

ZigBee devices connect to a central hub and form a mesh, so many devices (more than Bluetooth's seven) can be connected (1) and the signal can be relayed from device to device to reach every room (1).

ZigBee has a longer range (up to about 100 m vs about 10 m for Bluetooth) (1) and uses very little power, so battery sensors/switches last a long time (1).

QUICK CHECK — Section 5

1. Give three reasons why organisations still use Ethernet.

   Cheap; more secure than wireless; reliable (error detection); compatible; easy to install.

2. Describe the pairing process.

   Host scans for Bluetooth devices → list shown → user selects device → may enter/confirm PIN → connection created.

3. What is handover in a cellular network?

   Passing a phone's connection from one cell's base station to the next as the user moves.

4. Which method is linked to the Internet of Things?

   ZigBee.

SECTION 6 · SPEC 2.1.3 · TEXTBOOK PP.53–55

Network standards and protocols

> Sending a message across the world involves many jobs: turning it into data, splitting it up, addressing it, choosing a route, checking for errors and turning it into signals. Network **models** divide these jobs into **layers**, each with a specific function. You need to know the **OSI model**, **TCP/IP**, **VOIP** and **SIP**.

KEY TERMS

Protocol	A set of rules that govern the transfer of data.
OSI model	A theoretical framework that is the basis of many protocols; it divides network communication into seven layers.
TCP/IP model	A protocol suite (four layers) used to transmit data across the internet.
VOIP	Voice over Internet Protocol — used to send data when one person makes a voice call to another using the internet.
SIP	Session Initiation Protocol — a protocol used as part of VOIP to start, manage and end the call (session).

6.1 Why use layers?

- Each layer has **one job**, so the problem is broken into smaller, manageable parts.
- Layers can be **developed and changed independently** — e.g. switching from Wi-Fi to Ethernet (lower layers) does not change your web browser (top layer).
- **Different manufacturers** can build compatible hardware and software by following the standard for a layer.
- Makes it easier to **find faults** — engineers can test layer by layer.

6.2 The OSI model p.53–54

The **Open Systems Interconnection (OSI) model** was published in **1984** by the International Organization for Standardization (ISO). It is a **theoretical** framework: it divides the transmission of data from one computer to another into seven **layers**. Data is passed from one layer to the next. Many real protocols are based on some of its layers.

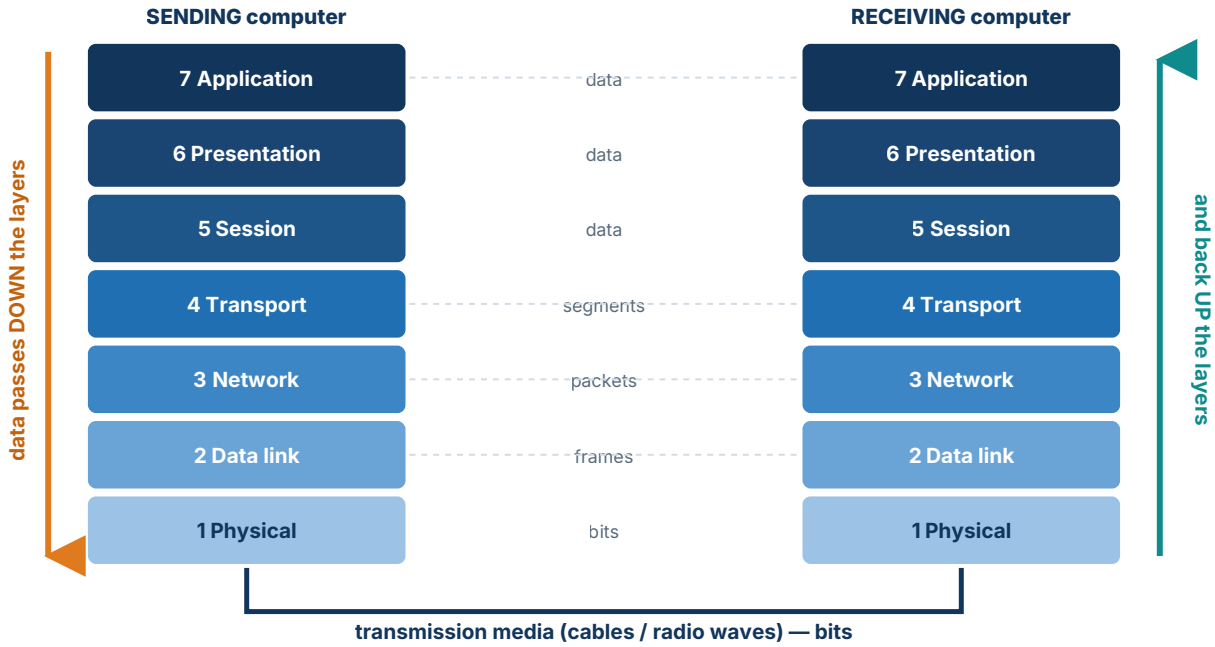


Figure 6.1 Data passes down the seven OSI layers on the sender and back up on the receiver. Grey labels show the unit of data at each layer.

TABLE 6.1 THE OSI MODEL (FROM TEXTBOOK TABLE 2.7)

Layer	What it does
7 Application	The **top** layer; directly interacts with the data from the user. It receives data from the user and displays data for the user. Applications such as a word processor are *not* part of it, but web browsers and email clients are. It provides several protocols used in transmission, including **HTTP** and **FTP**.
6 Presentation	Prepares data for use in the application layer — makes sure it is in the **correct format** for the intended device. May involve **encoding, encrypting or compressing** the data.
5 Session	Creates channels of communication called **sessions** between devices. Responsible for **opening** a session, keeping it open long enough to transmit the data, then **closing** it.
4 Transport	Breaks data down into **segments** to prepare it for transmission, and **reassembles** them afterwards. Sends data at a speed the network can handle. **Checks for errors** and requests that any data with errors be **re-sent**.
3 Network	Breaks segments down into smaller units called **packets**, and reassembles received packets into segments. Finds the **best route** for packets across networks — normally done using a **router** (IP addresses).
2 Data link	Similar to the network layer, but used when transmitting from one device to another **within a single network** (e.g. a LAN). Converts packets into **frames** and transmits them between devices. Two sublayers: **logical link control** — manages protocols and error checking; **media access control** — uses **MAC addresses** to connect devices.
1 Physical	All the **wired cables or wireless connections** between devices. Data is broken down into **bits** so it can be transmitted down a cable or across radio waves.

EXAM TIP — REMEMBER THE ORDER

From the top (7 → 1): **A**ll **P**eople **S**eem **T**o **N**eed **D**ata **P**rocessing = **A**pplication, **P**resentation, **S**ession, **T**ransport, **N**etwork, **D**ata link, **P**hysical. Make up your own:

e.g. (1 → 7) "Please Do Not Throw Sausage Pizza Away"

6.3 The TCP/IP model p.54–55

TCP/IP is the main protocol suite used by devices communicating on the **internet**. Each time data is sent from one device to another over the internet, TCP/IP is used to prepare and send it. It has **four layers** instead of seven, and it is **older** than the OSI model.

TABLE 6.2 THE TCP/IP MODEL (TEXTBOOK TABLE 2.8)

Layer	Description
Application	Does the job of the OSI **application, presentation and session** layers.
Transport	Same as the OSI transport layer. Uses the **Transmission Control Protocol (TCP)**.
Network / Internet	Same as the OSI network layer. Uses the **Internet Protocol (IP)**.
Link / Physical	Does the job of the OSI **data link and physical** layers.

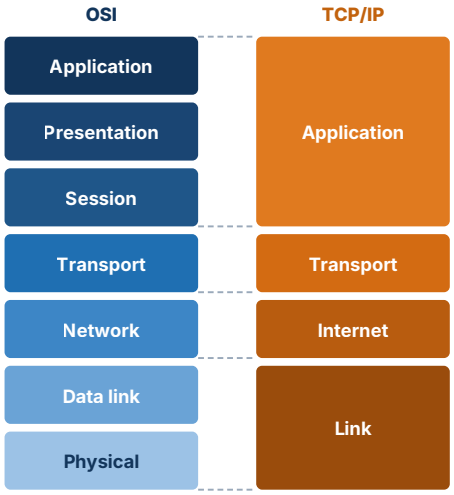


Figure 6.2 How the four TCP/IP layers map to OSI

Common protocols at the TCP/IP application layer (Textbook Activity 9):

Protocol	Used for	Protocol	Used for
HTTP / HTTPS	requesting and sending web pages (HTTPS is encrypted)	**DNS**	converting domain names (e.g. pearson.com) into IP addresses
FTP	transferring files between computers	**DHCP**	automatically giving devices an IP address
SMTP	sending email	**SSH**	secure remote log-in to another computer
POP3 / IMAP	receiving/downloading email (IMAP keeps it synchronised on the server)	**SIP**	setting up VOIP calls (see 6.4)

✓ FACT CHECK — TCP/IP

History: the textbook says TCP/IP "was developed by the US Department of Defense in the 1960s". More precisely: the DoD funded the ARPANET, which began in **1969** using an earlier protocol (NCP). TCP was designed by **Vint Cerf and Bob Kahn** in **1973–74** with DoD (DARPA) funding, and ARPANET switched to TCP/IP on **1 January 1983**. It is still true that TCP/IP is older than the OSI model (1984).
Transport layer: the textbook says it "uses TCP". It also uses **UDP** (User Datagram Protocol), which is faster but does not re-send lost data — it is used for live video, online games and the voice data in VOIP calls, where a late packet is useless anyway.

6.4 VOIP and SIP p.55

VOIP (Voice over Internet Protocol) is used to send data when a person makes a **voice call** to another person using the **internet** rather than the traditional telephone network. The first commercial VOIP software appeared in **1995**. It lets people call from devices such as a PC or laptop as well as phones. Apps such as Skype, FaceTime, WhatsApp calling and Teams use VOIP.

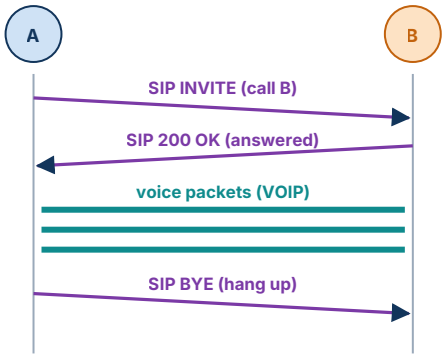


Figure 6.3 SIP sets up and ends the call; the voice travels as VOIP packets

VOIP is responsible for:

1. converting a person's **voice** (analogue) into **digital data**;
2. **compressing** it for transmission;
3. splitting it into packets and **transmitting** it across the internet.

VOIP uses another protocol, SIP (Session Initiation Protocol). SIP works at the **application layer**. It is a **signalling** protocol: it **starts** (initiates) the session by locating the other user and ringing them, **manages** it (e.g. adding a person, putting on hold) and **ends** it when someone hangs up. SIP does not carry the voice itself — the compressed voice packets are carried separately (normally by RTP over UDP).

Benefits of VOIP	Drawbacks of VOIP
Calls are cheap or free, even international ones	Needs a reliable internet connection — no internet, no calls
Can use PCs, laptops, tablets and phones; video and screen-sharing	Quality suffers with high **latency, jitter or packet loss**
Easy to add features (voicemail to email, conference calls)	Power cut can stop the router and so the calls; emergency-call location may be less reliable

EXERCISE 6.1 — Order and name the layers Recall

Layer	Name	Main function (one phrase)
7	Application	interacts with user data; HTTP, FTP
6	Presentation	formats, encodes, encrypts, compresses data
5	Session	opens, maintains, closes sessions

4	Transport	segments; error checking; re-send requests
3	Network	packets; best route; IP addresses; router
2	Data link	frames within one network; MAC addresses
1	Physical	cables/wireless; bits as signals

EXERCISE 6.2 — Which layer?

Apply

1. A file is encrypted before sending. → Presentation (6)
2. A router chooses the best path for a packet. → Network (3)
3. A segment arrives with an error and a re-send is requested. → Transport (4)
4. A switch uses a MAC address to send a frame to the right PC. → Data link (2)
5. Bits travel as light pulses along a fibre. → Physical (1)
6. A web browser sends an HTTP request. → Application (7)
7. A connection between two devices is opened, kept open and then closed. → Session (5)

EXERCISE 6.3 — Explain

4 marks each

1. Explain **two** differences between the OSI model and the TCP/IP model.

   OSI has seven layers, TCP/IP has four (1) — TCP/IP's application layer combines OSI's application, presentation and session layers, and its link layer combines data link and physical (1). OSI is a theoretical reference framework (1), while TCP/IP is the practical protocol suite actually used on the internet; TCP/IP is also older (1).

2. Explain the roles of VOIP and SIP when Leila makes a call to her grandmother using a laptop.

   SIP sets up the call — it locates/contacts the grandmother's device and starts the session (1), manages it and ends it when either hangs up (1). VOIP converts Leila's voice into digital data and compresses it (1), then sends it as packets across the internet to the grandmother's device, where it is converted back to sound (1).

EXERCISE 6.4 — Research

Textbook Activity 9

Use the internet to find out which protocols are used in the application layer of TCP/IP. Choose three that are *not* in the table above and describe them.

e.g. Telnet — unencrypted remote log-in; SNMP — monitoring/managing network devices; NTP — synchronising clocks; RDP — remote desktop; TLS is often listed alongside HTTPS to encrypt application data.

QUICK CHECK — Section 6

1. What does OSI stand for and when was it published?

   Open Systems Interconnection; 1984.

2. At which OSI layer are packets created and routed?

   Network layer (3).

3. Name the two sublayers of the data link layer.

   Logical link control (LLC) and media access control (MAC).

4. Which protocol at the TCP/IP internet layer deals with addressing?

   IP — Internet Protocol.

5. Give one reason a VOIP call might sound choppy.

   High jitter, latency or packet loss / low bandwidth on the connection.

SECTION 7 · SPEC 2.2.5 · TEXTBOOK PP.56–57

IP and MAC addressing

> Just as a letter needs an address, every device on a network needs addresses so that data reaches the right place. Each device has **two**: a **MAC address**, fixed in its network card, used to deliver data *within* a network; and an **IP address**, given when it connects, used to route data *between* networks.

KEY TERMS

IP address	A unique address given to a device when it connects to a network.
MAC address	A unique address given to the NIC in each device at the point of manufacture.
Static IP	The device always has the same IP address.
Dynamic IP	The device's IP address can change each time it connects to a network.
DHCP	Dynamic Host Configuration Protocol — automatically assigns IP addresses to devices when they join a network.
ISP	Internet service provider — the company that provides a user's internet connection; it can also assign IP addresses.

7.1 IP addresses p.56–57

An IP address is used by **routers** to send data packets to the correct network and device. It is placed in the header of every packet (source and destination). An IP address can be assigned by the network (usually the router, using DHCP) or by the **ISP**.

- Each address is **unique to the network** the device is connected to.
- It can be **static** or **dynamic**.
- There are two versions: **IPv4** and **IPv6**.

IPv4

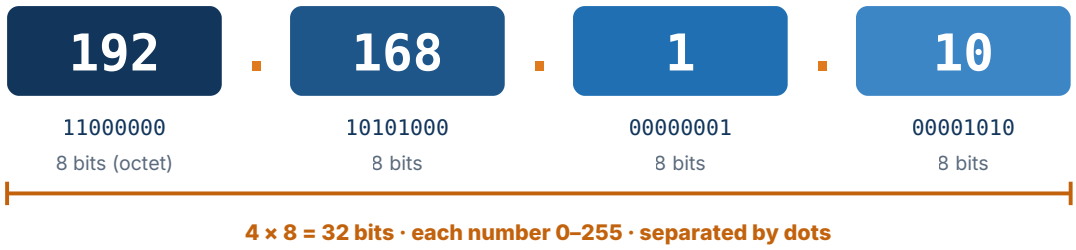


Figure 7.1 The structure of an IPv4 address, in denary and binary

32 bits gives $2^{32} \approx$ **4.3 billion** possible addresses — not enough for every phone, computer and IoT device in the world. This is the main reason IPv6 was created.

IPv6

IPv6 uses **128 bits**, giving $2^{128} \approx 3.4 \times 10^{38}$ addresses. It is written as **eight groups** of **four hexadecimal digits** (each group = 16 bits), separated by **colons**.

`AB01:EF01:2368:BB30:4324:10DB:0455:BA10` textbook example

TABLE 7.1 FEATURES OF IPV4 AND IPV6 (FROM TEXTBOOK TABLE 2.9)

IPv4	IPv6
32-bit address	128-bit address
Normally represented in **denary**	Normally represented in **hexadecimal**
Four sets of digits	Eight sets of digits
Each set separated by **dots**	Each set separated by **colons**
≈4.3 billion addresses	≈3.4×10^{38} addresses
—	Groups of zeros can be shortened (rules below)
e.g. `128.11.32.40`	e.g. `AB01:EF01:2368:BB30:4324:10DB:0455:BA10`

SHORTENING IPV6 ADDRESSES — THE TWO RULES (RFC 4291 / RFC 5952)

1. **Leading zeros** in any group can be left out: `0455` → `455`, `0000` → `0`.
2. **One** run of one or more consecutive all-zero groups can be replaced by a double colon `::`. The `::` may be used **only once** in an address — otherwise nobody could tell how many zero groups each one stands for.

Example: `2001:0DB8:0000:0000:0000:FF00:0042:8329` → `2001:DB8::FF00:42:8329`

✓ FACT CHECK — THE TEXTBOOK'S IPV6 EXAMPLE IS INVALID

The textbook says "a set of four 0 digits can be replaced by a double colon" and gives the example `AB10:F1B1::AF00:3121::8998:FF22`. That address uses `::` **twice**, which the IPv6 standard (RFC 4291) does not allow: it has six written groups, so two zero groups are missing — but we cannot tell whether they are 1 + 1, 2 + 0 or 0 + 2. Also, `::` can replace *one or more* zero groups, not only one. If both gaps were single zero groups, the correct form is `AB10:F1B1:0:AF00:3121:0:8998:FF22`. **In the exam:** state that `::` replaces consecutive groups of zeros and can be used only once.

Static and dynamic IP addresses

Static	Dynamic
Always the **same**	Can **change** each time the device connects
Set manually by a network manager (or reserved)	Given automatically by **DHCP**
Used for **servers, printers, routers**, CCTV — things others must always find	Used for most user devices: phones, laptops, PCs
Easier for remote access; can cost extra from an ISP	Easy to manage; addresses are reused when devices leave

DHCP runs on a server or (at home) on the router. It keeps a **pool** of available addresses and **leases** one to each device that joins, for a set time. The exchange has four steps — remember **DORA**: the device **D**iscovers a server, the server **O**ffers an address, the device **R**equests it, and the server **A**cknowledges, confirming the lease.

When the lease runs out, the device must renew it or the address goes back into the pool for another device.

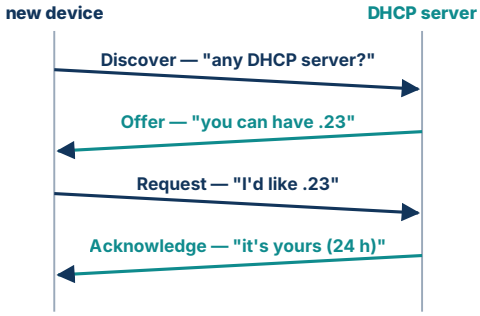


Figure 7.2 DHCP: Discover, Offer, Request, Acknowledge

Public and private addresses

Devices inside a home or school LAN normally have **private** IP addresses, which only need to be unique within that LAN — e.g. `192.168.x.x`, `10.x.x.x` or `172.16.x.x – 172.31.x.x`. The router has one **public** IP address (from the ISP) that is unique on the internet, and translates between them (NAT). This is why the textbook says an IP address is "unique to the network that the device is connected to", and why many devices around the world can share the same private address.

7.2 MAC addresses

A **media access control (MAC)** address is a unique identifier given to every **NIC** when it is **manufactured**. It is used by **switches and bridges** (OSI data link layer) to deliver frames to the correct device **within a LAN**.

- **48 bits**, written as **12 hexadecimal digits** in six pairs, separated by colons or hyphens.
- The first half identifies the **manufacturer**; the second half is the device's **serial number** from that manufacturer.
- It normally **does not change** — although it can be altered in software ("spoofed"), and many phones now use a random MAC on Wi-Fi for privacy.

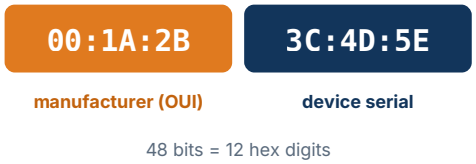


Figure 7.3 Structure of a MAC address

TABLE 7.2 IP ADDRESS VS MAC ADDRESS

	IP address	MAC address
Assigned	when the device connects (DHCP, ISP or manually)	when the NIC is manufactured
Can change?	yes (dynamic) — changes with network	normally fixed
Size / format	IPv4 32-bit denary; IPv6 128-bit hex	48-bit, 12 hex digits
Used by	routers — between networks	switches, bridges — within a LAN
OSI layer	3 Network	2 Data link
Analogy	your postal address (changes if you move)	your passport number (stays with you)

EXERCISE 7.1 — Valid or not?

Apply

Tick if valid. If invalid, give the reason.

Address	Valid?	Reason if invalid
192.168.1.256	×	256 is above 255 (max for 8 bits)
10.0.0.1	✓	—
172.16.4	×	only three sets of numbers — needs four
2001:DB8::1	✓	—
FE80::1::5	×	:: used twice
2001:0DB8:85G3:0000:0000:8A2E:0370:7334	×	G is not a hexadecimal digit
00:1A:2B:3C:4D:5E	✓ (MAC)	valid MAC address, not an IP address

EXERCISE 7.2 — Shorten and expand IPv6

Apply

1. Shorten `2001:0DB8:0000:0000:0000:0000:1428:57AB`

   2001:DB8::1428:57AB

2. Shorten `FE80:0000:0000:0000:0202:B3FF:FE1E:8329`

   FE80::202:B3FF:FE1E:8329

3. Expand to full form `2001:DB8::1`

   2001:0DB8:0000:0000:0000:0000:0000:0001

4. Shorten `2001:0000:0000:ABCD:0000:0000:0000:0001` (careful — choose the longer run)

   2001:0:0:ABCD::1

EXERCISE 7.3 — Convert IPv4 to binary

Number skills

Convert each part to 8-bit binary. Use the column headings 128 64 32 16 8 4 2 1.

Address	Octet 1	Octet 2	Octet 3	Octet 4
128.11.32.40	10000000	00001011	00100000	00101000
172.16.254.3	10101100	00010000	11111110	00000011

Why can no part of an IPv4 address be greater than 255?

Each part is 8 bits; the largest 8-bit number is 11111111 = 255.

EXERCISE 7.4 — Find your addresses

Textbook Activity 11 · Investigate

Find the IP address of the computer you are using (Windows: open Command Prompt and type `ipconfig /all`; macOS: System Settings → Network → Details; phone: Settings → Wi-Fi → (i)/ details).

IPv4 address:	IPv6 address:
MAC (physical) address:	Default gateway (router):
DHCP enabled? (yes/no)	DNS server:

Does your IP address look like IPv4 or IPv6? How do you know? Is it public or private?

Most likely an IPv4 address such as 192.168.x.x — four denary numbers separated by dots (IPv6 would be hex groups separated by colons). 192.168.x.x/10.x.x.x is a private address given by the router's DHCP; the public address belongs to the router.

QUICK CHECK — Section 7

1. State **three** differences between IPv4 and IPv6.

   32-bit vs 128-bit; denary vs hexadecimal; 4 sets vs 8 sets; dots vs colons; ~4.3 billion vs ~3.4×10^{38} addresses.

2. Why was IPv6 needed?

   IPv4 (≈4.3 billion addresses) was running out as the number of connected devices grew.

3. What does DHCP do?

   Automatically assigns (leases) a dynamic IP address to each device that joins the network.

4. Why would a web server have a static IP address?

   So its address never changes and clients/DNS can always find it.

5. Which device uses MAC addresses to forward data?

   A switch (also a bridge).

SECTION 8 · SPEC 2.2.4 · TEXTBOOK P.53 ACTIVITY 8

Network design

> The specification asks you to "produce outline designs for networks to meet specified requirements that take account of location of devices". In the exam this usually means **drawing a labelled diagram**. Marks are given for each correct **connection** — the right device joined to the right device with the right **medium/protocol** written on the link.

8.1 A step-by-step method

Step	Do this	Ask yourself
1	**List every requirement** — underline each one in the question.	Which devices? Which must be wired / wireless? Any named protocol (ZigBee, powerline)?
2	**Map the locations**.	Which room/floor is each device in? Distances? Walls? Is a device mobile or fixed?
3	**Choose the central devices**.	Router/modem for internet access; switch for several wired devices; WAP for wireless; hub for ZigBee devices.
4	**Choose the medium for each link**.	Ethernet for fixed, fast, secure; Wi-Fi for mobile; powerline between floors; Bluetooth for close peripherals; ZigBee for smart devices; fibre for long distance.
5	**Draw and label**.	Every device named; every line labelled with the medium/protocol; show rooms/floors with a dashed boundary if location matters.
6	**Tick off** every requirement on your diagram.	Have I met all of them? Have I added anything that wasn't asked for?

EXAM TIP — SYMBOLS

You do **not** need special symbols. A labelled box ("router", "switch", "PC 1") is fine. Use a **solid line** for wired and a **dashed or zig-zag line** for wireless, and **write the protocol on the line** (e.g. "Ethernet", "Wi-Fi", "ZigBee", "powerline"). Include a small key.

8.2 Things to consider about location

- **Distance** — Ethernet copper runs ≈100 m; Bluetooth ≈10 m; Wi-Fi range falls through walls/floors.
- **Obstacles** — thick walls, metal, floors weaken Wi-Fi; infrared needs line of sight.
- **Mobility** — devices that move need wireless.
- **Where the router is** — usually where the internet line enters the building.
- **Placing WAPs** — central and high, to cover the area; add WAPs/repeaters for weak spots.
- **Security** — servers and switches in a locked room; wired for sensitive devices.
- **Power** — devices need sockets; powerline needs same circuit.
- **Interference** — 2.4 GHz is crowded (Wi-Fi, Bluetooth, ZigBee, microwave ovens).

8.3 Worked example — a small dental surgery

Requirements: a receptionist PC and a printer in reception; two treatment rooms, each with a PC that must be wired for security; a file server holding patient records in a locked comms cupboard; free Wi-Fi for patients in the waiting room that cannot access the patient records; internet access for all.

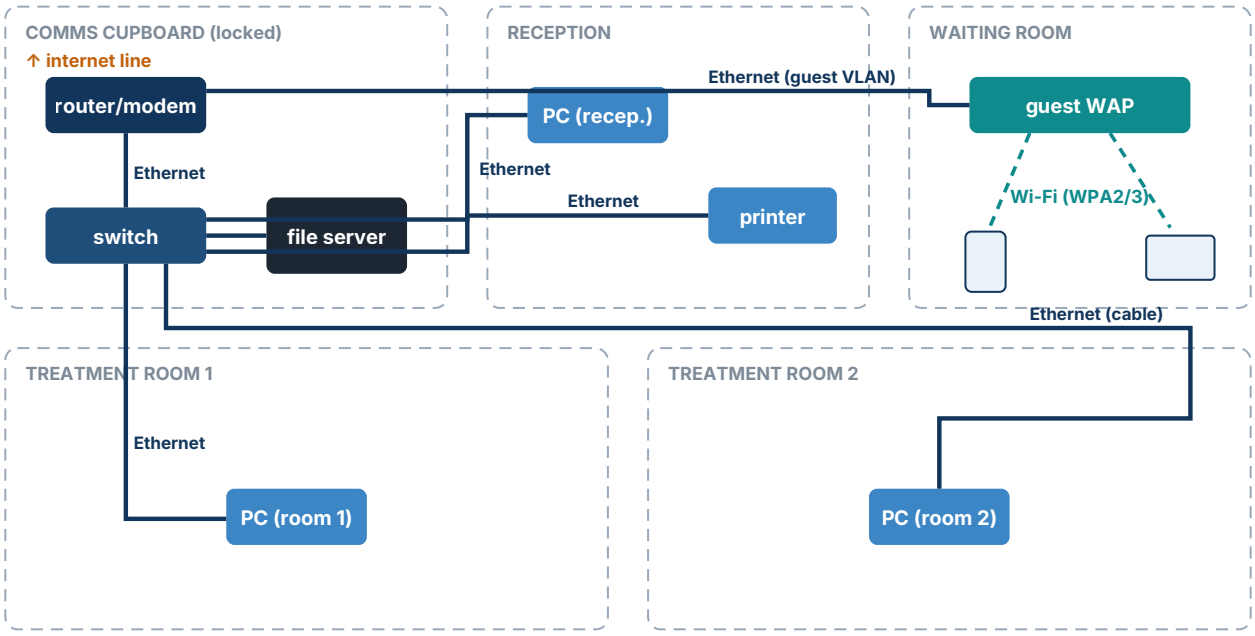


Figure 8.1 Outline design for the dental surgery — every link labelled; patient Wi-Fi kept on a separate guest network

Why these choices? Wired Ethernet for the treatment-room PCs (requirement: security; fixed devices). Server and switch in the locked cupboard (physical security). A separate guest WAP/network so patients can't reach patient records. The printer is wired, but could be wireless if near the reception PC.

EXERCISE 8.1 — Tariq's home network

Textbook Activity 8 · Design

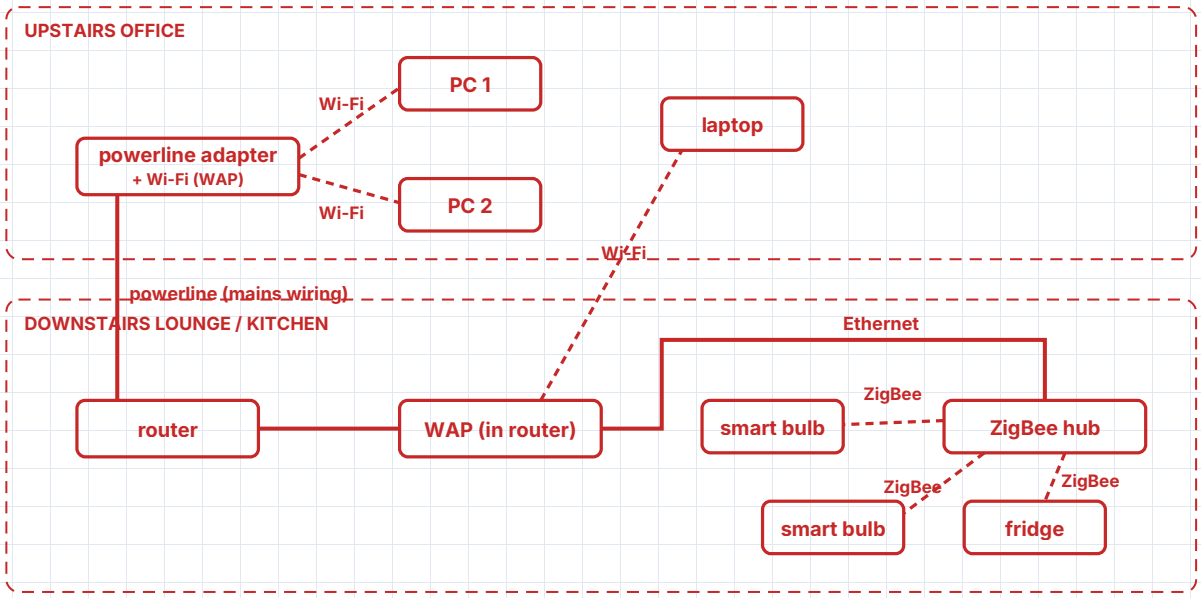
Tariq has a home network and wants to add some new smart devices. Tariq wants:

- the **two existing PCs** in the upstairs office to remain connected using a **wireless** connection
- the existing **laptop** to be connected using **Wi-Fi**
- a **WAP** to allow any new devices that have wireless capabilities to be connected
- the new **smart bulbs** to be connected using **ZigBee**
- the new **smart refrigerator** to be connected using **ZigBee**
- a **powerline** connection between the **router in the downstairs lounge** and the PCs in the upstairs office.

Step 1. Underline the devices and the protocol for each. **Step 2.** Notice the possible contradiction: the PCs are "wireless" *and* reached by powerline. How can both be true? Write your idea:

Powerline carries the connection from the downstairs router to the upstairs office; there, the second powerline adapter has a built-in Wi-Fi access point (or connects to a WAP), and the two PCs connect to it wirelessly. So the long link is powerline, the last few metres are Wi-Fi.

Step 3. Draw the network. Show downstairs and upstairs, and label every link.



Model answer: powerline adapters link router and office; Wi-Fi from the upstairs adapter/WAP to PC 1 and PC 2; laptop on Wi-Fi via the router's WAP; ZigBee hub (wired to router) with ZigBee links to both bulbs and fridge. Accept the hub built into a smart speaker.

Step 4. Tick each requirement your diagram meets: ☐ PCs wireless ☐ laptop Wi-Fi ☐ WAP ☐ bulbs ZigBee ☐ fridge ZigBee ☐ powerline router ↔ office

EXERCISE 8.2 — Design a school library network

Design · 10 marks style

A school library has: 12 student desktop PCs in a row along one wall; a librarian's PC at the issue desk which accesses the school's management system server in the main building 400 m away; a networked printer; tablets that students borrow and use anywhere in the library; a barcode scanner connected to the librarian's PC; smart lights controlled from a wall panel. Design the network.

Plan: for each device, write the medium/protocol you will use.

Device / link	Medium / protocol	Reason
12 student PCs → switch	Ethernet	fixed, fast, cheap, secure
Library switch → main building (400 m)	fibre optic	over 100 m — copper too far; high bandwidth
Tablets	Wi-Fi via WAP(s)	mobile anywhere in library
Barcode scanner → librarian PC	Bluetooth or USB	short range peripheral
Printer	Ethernet (or Wi-Fi)	shared by all; fixed
Smart lights → wall panel/hub	ZigBee	low-power IoT devices; hub

Now draw your design.

Model: student PCs + librarian PC + printer → switch (Ethernet); switch → fibre link → main building switch/server; WAP(s) on switch (Ethernet) → tablets (Wi-Fi); scanner → librarian PC (Bluetooth/USB); smart lights → ZigBee hub/panel. Award 1 mark per correctly labelled connection.

EXERCISE 8.3 — Critique a design

Evaluate

A student designed a network for a three-floor office: one Wi-Fi router on the ground floor for all 60 staff laptops, hubs to connect the 20 desktop PCs, and Bluetooth to connect a printer on the second floor. Identify **three** problems and suggest improvements.

1. One router on the ground floor won't give Wi-Fi coverage to three floors and 60 users would congest it → add WAPs on each floor (wired back to a switch).

2. Hubs send data to every device → many collisions/poor performance → use switches.

3. Bluetooth range is ≈10 m and the printer is two floors away → connect the printer by Ethernet (or Wi-Fi) to the network.

(Also: no server/backup mentioned; no firewall.)

SECTION 9 · SPEC 2.3.1–2.3.2 · TEXTBOOK PP.57–62

Network security

> When a device is connected to a network, the device *and all the data stored on it* become **vulnerable**. Data can be valuable, and some people will try to steal it or corrupt it. In this section you will learn **why** people attack networks, **what** the threats are, the **impact** on individuals and organisations, and the **hardware and software solutions** used to secure a network.

9.1 Why do people attack networks? p.57

Motive	What it means / example
Fraud	Stealing money — e.g. using stolen card or bank details to make purchases or transfers.
Identity theft	Stealing personal details (name, date of birth, ID numbers) to pretend to be someone, e.g. to take out loans in their name.
Intellectual property theft	Stealing designs, source code, research or creative work to copy or sell.
Revenge	A disgruntled ex-employee deletes data or leaks information.
Ethical protests	"Hacktivists" deface websites or leak data to protest against an organisation's actions.
Entertainment value	Hacking for fun, a challenge or to show off.

9.2 The threats

TABLE 9.1 COMMON NETWORK THREATS

Threat	How it works
Hacking	Gaining **unauthorised access** to a computer system or network — e.g. by guessing weak passwords or exploiting unpatched software — to steal, change or delete data.
Malware	**Malicious software** designed to damage a device or the data on it. Usually installed when a user unknowingly downloads an infected file. • **Virus** — attaches to a file; spreads when the file is run/shared; damages or deletes data. • **Worm** — copies itself across a network without user action, using up bandwidth. • **Trojan horse** — disguised as useful software; secretly opens a "back door". • **Spyware** (incl. keyloggers) — secretly records activity such as passwords and sends it to the attacker. • **Ransomware** — encrypts the victim's files and demands payment to unlock them.
Phishing	Fake emails, texts or websites that look genuine, tricking users into giving passwords or bank details or clicking a malicious link.
Denial of service (DoS / DDoS)	Flooding a server or network with so much traffic that genuine users cannot use it. A **distributed** DoS uses thousands of infected devices (a botnet). Reduces **availability**.
Interception (eavesdropping)	Capturing data as it travels across the network — especially easy on wireless/open networks. "Man-in-the-middle" attacks secretly relay and alter messages.

Social engineering	Manipulating **people** rather than technology — e.g. phoning staff pretending to be IT support to get a password; "shoulder surfing".
Insider threats & accidents	Staff misusing access, or accidentally deleting data, losing a laptop or sending data to the wrong person.

Open networks

An **open network** is one that anyone can join, usually **public Wi-Fi with no password** (cafés, airports, hotels). Risks:

- Data is often sent **unencrypted**, so others on the network can intercept it.
- Attackers can set up a fake hotspot with a similar name (an "evil twin").
- Other users on the same network can try to access your device.

STAYING SAFE ON OPEN NETWORKS

- Only use **HTTPS** websites; avoid banking.
- Use a **VPN** to encrypt all traffic.
- Turn off file sharing; keep the firewall on.
- Use **tethering** to your own phone instead.
- Check the exact network name with staff.

9.3 Impact on individuals and organisations

Individuals	Organisations
• Money stolen from bank accounts; fraudulent purchases • Identity theft — damaged credit rating, loans in their name • Loss of personal files and photos (e.g. ransomware) • Embarrassment, stress, blackmail if private data is leaked • Time and cost to recover accounts and devices	• **Financial loss** — theft, ransom, cost of recovery, lost sales during downtime • **Reputational damage** — customers lose trust and go elsewhere • **Legal consequences** — fines under data-protection laws (e.g. GDPR) if personal data is not protected • Loss of intellectual property to competitors • Operations stop — e.g. hospitals cancelling appointments

9.4 How to secure a network p.58–62

The specification lists seven methods. Most networks use several together — "defence in depth".

a. Firewall

A firewall sits between a network (or device) and the outside world, usually the internet. It **monitors all incoming and outgoing traffic** and **checks each packet against a set of rules** (e.g. allowed IP addresses, ports, protocols). Traffic that breaks the rules is **blocked**. It can be **hardware** (a device or part of the router) or **software** (e.g. built into the operating system).

It helps stop hackers gaining access and can stop malware on a device "phoning home". It *cannot* stop a user willingly opening an infected email attachment.

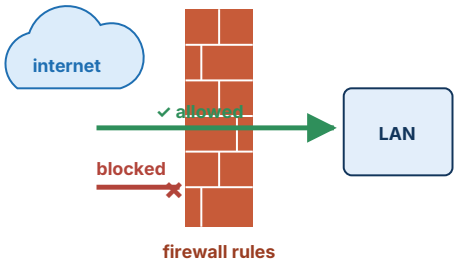


Figure 9.1 A firewall filters traffic by rules

b. Security settings p.59

Reviewing the **security settings** of a device or account — the ways a user can control who can see or access data they store. For example, on a social media account you can make more of your data **private** so that other users cannot see it, and turn on **two-factor authentication**. On a router: change the default admin password, use WPA3/WPA2 encryption, turn off WPS, update the firmware, hide or rename the SSID.

c. Anti-malware p.59

Anti-malware is software that **scans** a device for malware. If it finds a file it thinks is malware, it can **delete** it or **quarantine** it (isolate it so it can't run) for the user to review. Some can scan a file **before** it is downloaded. It must be **updated regularly**, because new malware is created all the time and updates tell the software about new known threats.

d. User controls, access rights and profiles p.59

Access rights are restrictions on who can see and edit data. They are **not** designed to stop a hacker getting into an account; they **limit the data** that the hacker — or any user — can see. They are attached to a person's username (their **user profile**): when the user logs in, the server knows what data they can see and what they can do with it (**read only**, **read and edit**, no access).

Example: in a school the head teacher can **amend** any student's data; other teachers can only **see** it. This keeps data more secure: it is less likely to be accidentally deleted or corrupted, and a hacker who gets into one teacher's account cannot see everything.

User	Student records	Staff salaries	Exam papers
Head teacher	read / edit	read	read / edit
Class teacher	read	no access	read (own subject)
Finance officer	no access	read / edit	no access
Student	own record (read)	no access	no access

e. Authentication — passwords, two-factor, biometrics

Authentication checks that a user really is who they claim to be. There are three types of factor:

SOMETHING YOU KNOW
password, PIN, security question

SOMETHING YOU HAVE
phone (code by SMS/app), security key, swipe card

SOMETHING YOU ARE
fingerprint, face, iris, voice (biometrics)

- **Passwords** — should be long, a mix of character types, not a dictionary word, not reused; stored by the system as a hash, not plain text. Weakness: can be guessed, stolen by phishing or keyloggers.
- **Two-factor authentication (2FA)** — uses two *different* types of factor, e.g. password + code sent to phone. A stolen password alone is not enough.
- **Biometrics** — use data such as facial features, a fingerprint or a voice as a password. Hard to forget or share; but expensive hardware, can be fooled, raises privacy concerns, and unlike a password it cannot be changed if stolen.

f. Encryption

Encryption scrambles data (**plaintext**) into an unreadable form (**ciphertext**) using an algorithm and a **key**. Only someone with the correct key can **decrypt** it. Encryption does **not** stop data being intercepted — it makes intercepted data **useless**.

- **Symmetric** — the same key encrypts and decrypts. Fast, but the key must be shared safely.
- **Asymmetric (public key)** — a **public key** encrypts; only the matching **private key** decrypts. Used to set up secure connections such as HTTPS.
- Uses: HTTPS/TLS websites, WPA2/WPA3 Wi-Fi, VPNs, encrypted laptops/USB drives, messaging apps.

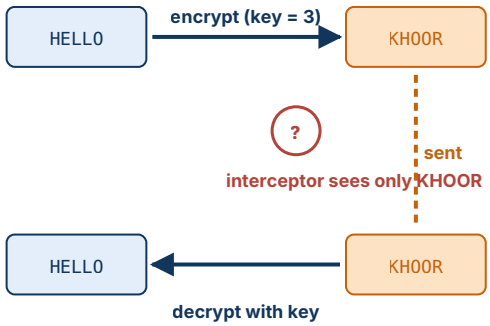


Figure 9.2 A simple shift (Caesar) cipher — real encryption uses far stronger algorithms

g. Physical controls

Protecting the **hardware** itself so that no one can steal it or plug into it:

- servers, switches and routers in a **locked room/cabinet**; key cards or biometric door locks; visitor sign-in
- **CCTV** and security guards; alarms
- cable locks for laptops; disabling unused network ports and USB ports
- protecting against fire/flood/power loss: UPS (backup power), fire suppression, off-site backups

TABLE 9.2 MATCHING THREATS TO SOLUTIONS

Threat	Main solutions
Hacking	firewall, strong passwords + 2FA, access rights, security updates
Malware	anti-malware (updated), firewall, user training, not running unknown files, backups (ransomware)
Phishing	user training, email filtering, 2FA (stolen password not enough)
Interception	encryption (WPA3, HTTPS, VPN), wired connections for sensitive data
DoS / DDoS	firewall rules, traffic filtering services, scalable servers
Theft of hardware	physical controls, encrypted drives
Insider misuse	access rights and user profiles, audit logs

EXERCISE 9.1 — Fill in the blanks

Recall

1. Anti-malware can delete a suspected file or quarantine it for the user to review. It must be updated regularly because new malware is created all the time.
2. Access rights limit the data a user (or hacker) can see and what they can do with it, e.g. read only or read and edit.
3. Two-factor authentication uses two different types of factor, e.g. a password and a code sent to a phone.
4. Encryption turns plaintext into ciphertext using a key.

5. A firewall checks incoming and outgoing traffic against a set of rules.
6. Ransomware encrypts a victim's files and demands payment.

EXERCISE 9.2 — Caesar cipher

Practise encryption

Using a shift of 3 (A→D, B→E … X→A, Y→B, Z→C):

```
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
D E F G H I J K L M N O P Q R S T U V W X Y Z A B C
```

1. Encrypt `NETWORK` → QHWZRUN
2. Decrypt `ILUHZDOO` → FIREWALL
3. Why would this cipher be easy for a hacker to break?

Only 25 possible shifts — a computer (or person) can try them all in seconds (brute force); letter patterns/frequencies are also kept.

EXERCISE 9.3 — Scenarios

Apply

Scenario	Threat	Best security measure
An email "from your bank" asks you to click a link and confirm your password.	phishing	user training; check sender; 2FA
A company's website is flooded with millions of requests and goes offline.	DDoS	firewall/traffic filtering; scalable servers
A student uses airport Wi-Fi and someone captures her login details.	interception (open network)	VPN / HTTPS / tethering
A laptop containing customer data is stolen from a car.	physical theft	encrypt the hard drive; strong log-in
A new receptionist can see every employee's salary.	insider / poor access rights	access rights / user profiles
All files on the office server are locked and a message demands Bitcoin.	ransomware	anti-malware; regular off-site backups

EXERCISE 9.4 — Explain

4 marks each

1. Explain why access rights improve the security of data in a school network.

Each user only sees/edits the data they need for their job (1), so data is less likely to be accidentally deleted or corrupted by someone who shouldn't change it (1). If a hacker gets into one account (1), they can only see the data that account is allowed to see, not all the organisation's data (1).

2. Explain why two-factor authentication is more secure than a password alone.

2FA needs two different types of factor, e.g. something you know (password) and something you have (phone) (1). If the password is stolen by phishing/keylogger/guessing (1), the attacker still cannot log in (1) without also having the user's phone/code, which changes each time (1).

3. Give two benefits and two drawbacks of biometric authentication.

Benefits: cannot be forgotten or lost; hard to share/copy; fast to use. Drawbacks: expensive hardware; can give false rejections/acceptances (e.g. dirty finger); privacy concerns over storing biometric data; cannot be changed if stolen.

EXERCISE 9.5 — Security plan

Discuss · 8 marks

A small accountancy firm stores clients' financial records on a file server. Staff work in the office and sometimes from home or from cafés. Discuss the security measures the firm should use.

Firewall between the office network and the internet to block unauthorised access. Anti-malware on every device, updated automatically, to detect/quarantine malware (especially ransomware). Strong passwords + 2FA for all accounts — financial data is a prime phishing target. Access rights: staff only see the clients they work on; only partners can delete. Encryption: encrypted laptops in case of theft; VPN when working from home or cafés because public Wi-Fi is open/unencrypted; WPA3 on office Wi-Fi. Physical: server in a locked room; CCTV. Regular encrypted off-site backups to recover from ransomware or fire. Staff training on phishing. Consider cost vs risk: losing client financial data would cause fines, reputational damage and loss of clients, so investment is justified.

Levels guidance: L3 (7–8) a range of measures, each explained AND applied (financial data, remote work/ cafés), with some weighing of cost/usability. L2 (4–6) several measures with some explanation. L1 (1–3) list of measures.

EXERCISE 9.6 — Check your own settings

Investigate

Review the security settings on one of your own accounts (social media, email or gaming). Record:

Is 2FA available? Is it on?	Who can see your posts / profile?
Which devices are logged in?	One setting you changed and why:

QUICK CHECK — Section 9

1. List four reasons why people attack networks.

   Fraud, identity theft, intellectual property theft, revenge, ethical protest, entertainment.

2. What is the difference between a virus and a worm?

   A virus attaches to a file and spreads when the file is run/shared; a worm replicates itself across a network on its own without user action.

3. Why must anti-malware be updated regularly?

   New malware is created all the time; updates add the latest known threats so they can be detected.

4. Give two physical security controls.

   Locked server room; key-card/biometric door locks; CCTV; cable locks; UPS.

5. Why are open networks a risk?

   Anyone can join; data is often unencrypted so it can be intercepted; fake hotspots.

SECTION 10 · MIXED TOPICS

Exam practice

> These questions are written in the style of the Unit 1 paper, which uses multiple-choice, short-answer, open-response and extended open-response questions (2 hours, 80 marks). Aim for about **1 mark per 1.5 minutes**. Total for this section: **59 marks** — about 90 minutes.

COMMAND WORDS

State / Give / Identify	A short answer — a word, phrase or fact. No explanation needed.
Describe	Say what something is like or what happens — the features, the steps.
Explain	Give a point **and** a reason/justification ("...because...", "...so that...").
Discuss	Explore different aspects of an issue — advantages and disadvantages, how factors interrelate.
Evaluate / Assess	Weigh up the points **and reach a supported judgement/conclusion**.

Part A — Multiple choice

8 marks

Tick **one** box for each question.

1. Which network component reads the MAC address of a data packet and sends it only to the destination device? **(1)**

   ☐ A Hub

   ☐ B Modem

   ☐ C Switch

   ☐ D Repeater

2. Which transmission medium is **not** affected by electromagnetic interference? **(1)**

   ☐ A Coaxial cable

   ☐ B Fibre optic cable

   ☐ C Powerline

   ☐ D Twisted pair cable

3. What is the variation in the time taken for packets to arrive called? **(1)**

   ☐ A Bandwidth

   ☐ B Error rate

   ☐ C Jitter

   ☐ D Throughput

4. At which layer of the OSI model are segments created? **(1)**
   - ☐ A Session
   - ☐ B Transport
   - ☐ C Network
   - ☐ D Data link

5. How many bits are in an IPv6 address? **(1)**
   - ☐ A 32
   - ☐ B 48
   - ☐ C 64
   - ☐ D 128

6. Which protocol is mainly used to connect low-power smart home devices through a hub? **(1)**
   - ☐ A Bluetooth
   - ☐ B Ethernet
   - ☐ C Infrared
   - ☐ D ZigBee

7. Which protocol automatically gives a device an IP address when it joins a network? **(1)**
   - ☐ A DHCP
   - ☐ B FTP
   - ☐ C SIP
   - ☐ D SMTP

8. Which of the following is a valid IPv6 address? **(1)**
   - ☐ A 2001:DB8::25::1
   - ☐ B 2001:DB8::25:1
   - ☐ C 2001.DB8.25.1
   - ☐ D 2001:DG8::25:1

Answers: 1 C · 2 B · 3 C · 4 B · 5 D · 6 D · 7 A · 8 B (A uses :: twice; C uses dots; D contains G, not hex).

Part B — Short and open response

33 marks

9. A new office is being set up.

(a) State the role of a **router** in the office network. **(2)**

Forwards data packets between networks (the office LAN and the internet) (1) by reading the destination IP address and using a routing table to choose the best route (1).

(b) Describe how a **switch** reduces data collisions compared with a hub. **(3)**

The switch stores the MAC addresses of connected devices in a table (1). It reads the destination MAC address of each packet (1) and forwards it only to that device's port instead of to all devices (1), so there is less traffic and fewer collisions.

10. Explain **two** reasons why a company would use fibre optic cable to connect two buildings 1.5 km apart. **(4)**

Fibre transmits light, which loses little strength over long distances (1), so it can cover 1.5 km where twisted pair is limited to about 100 m (1). It has very high bandwidth (1), so large volumes of data can be transferred quickly between buildings (1). / It is not affected by EMI (1) so data is reliable/fewer errors (1).

11. A download of 900 MB takes 3 minutes. Calculate the throughput in Mbps. Show your working. **(3)**

900 × 8 = 7 200 Mb (1); 3 min = 180 s (1); 7 200 ÷ 180 = 40 Mbps (1).

12. Compare IPv4 and IPv6 addresses. **(4)**

IPv4 is 32 bits but IPv6 is 128 bits (1); IPv4 is written in denary, IPv6 in hexadecimal (1); IPv4 has four sets separated by dots, IPv6 eight sets separated by colons (1); IPv6 provides far more addresses (≈3.4×10^{38} vs ≈4.3 billion), solving the shortage of IPv4 addresses (1). Max 4. Must be comparative.

13. Aisha uses a VOIP application to call her cousin abroad. Describe the role of SIP in this call. **(3)**

SIP is an application-layer signalling protocol (1) that initiates the session — locating/ringing the cousin's device (1) — manages it during the call and ends/terminates it when one person hangs up (1).

14. Aisha's call keeps breaking up. Explain **two** network metrics that could be causing this. **(4)**

High jitter (1): packets arrive at uneven intervals so speech sounds choppy/robotic (1). High packet loss (1): some voice packets never arrive, so words are missing (1). / High latency: delay causes people to talk over each other.

15. Give **two** benefits and **two** drawbacks of a peer-to-peer network compared with client–server. **(4)**

Benefits: cheaper to set up — no server (1); no specialist technician needed (1); one peer failing doesn't stop the network (1). Drawbacks: less secure — each user responsible for security (1); each user must back up their own data (1); slows down with many peers (1).

16. Explain why a network manager would give the school's file server a static IP address. **(2)**

Its IP address will never change (1), so all client computers can always find/connect to the server at the same address (1).

17. Explain how anti-malware protects a computer. **(4)**

It scans files/the device for malware (1), comparing them with known malware signatures/behaviour (1). Files found to be malware are deleted or quarantined (1). It can scan downloads before they are opened, and must be updated regularly to recognise new malware (1).

> **Marking note for Part B:** accept any valid alternative points with equivalent meaning; "explain" questions need the linked reason for the second mark in each pair.

Part C — Network design

10 marks

18. Mrs Silva runs a home business. She wants:

- a desktop PC and a laser printer in her office, next to the router, connected by cable
- her laptop and phone to connect anywhere in the house
- a smart thermostat and three smart plugs controlled from her phone
- wireless headphones for video calls on the laptop
- a security camera in the garden shed, 30 m from the house, where Wi-Fi does not reach
- internet access for all devices.

Draw a labelled diagram of a suitable network. Label each connection with the transmission method/protocol. **(10)**

Mark scheme — 1 mark per correctly labelled connection (max 10): router–internet (broadband/fibre/modem) · PC–router Ethernet · printer–router Ethernet · laptop–router Wi-Fi · phone–router Wi-Fi · ZigBee hub connected to router (Ethernet/Wi-Fi) · thermostat–hub ZigBee · smart plugs–hub ZigBee · headphones–laptop Bluetooth · shed camera via Ethernet cable / outdoor WAP / powerline (if same circuit) / Wi-Fi repeater — any justified. Accept smart plugs on Wi-Fi if labelled.

Part D — Extended response

8 marks

19. A chain of five clinics wants to let patients use free Wi-Fi in the waiting rooms. Clinic staff also use the network to access patient records stored on a central server.

Evaluate the security risks of providing open Wi-Fi to patients and the measures the clinics should take. **(8)**

Risks: open (password-less) Wi-Fi lets anyone in range join; unencrypted traffic can be intercepted, so if staff used the same network, patient data could be captured; attackers could try to access the server or spread malware to staff PCs; fake "evil twin" hotspots could trick patients; reputational damage and legal fines if medical data is leaked.

Measures: keep the patient Wi-Fi on a completely separate guest network/VLAN with no access to the staff LAN; firewall between them and the internet; use WPA2/WPA3 even for guests (password shown in reception) or a captive portal; staff network wired/encrypted with strong authentication and 2FA; access rights so staff only see needed records; anti-malware on staff devices; server physically secured; logging/monitoring.

Judgement: providing patient Wi-Fi is acceptable and improves the patient experience, provided it is fully separated from the clinical network; the main risk is to the clinic's data, not the Wi-Fi itself, so separation + firewall are the essential measures; costs are small compared with the impact of a breach of medical records.

Levels: L1 (1–2) isolated knowledge of risks or measures. L2 (3–5) risks and measures explained with some application to clinics; limited evaluation. L3 (6–8) well-applied analysis of risks and measures linked to medical data/open Wi-Fi, leading to a supported, balanced conclusion.

MY SCORE

Part A / 8	Part B / 33	Part C / 10	Part D / 8	Total / 59

Topics I lost marks on, and what I will revise:

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Revision checklist

> Rate yourself honestly for each statement: **R** = I can't do this yet, **A** = I'm unsure, **G** = I'm confident. Revisit every R and A before the exam, then rate yourself again.

Spec	I can…	R	A	G	Re-test
2.2.3	explain the role of NIC, hub, switch, router, modem, gateway, bridge, WAP, repeater, server and multi-function device				
2.2.3	explain why a switch performs better than a hub, and which address each device uses				
2.2.1	describe twisted pair, coaxial, fibre optic and powerline, including how each transmits data				
2.2.1	describe microwave, radio, light (infrared) and satellite transmission				
2.2.1	choose and justify a medium for a given situation				
2.1.1	describe client–server, peer-to-peer, ad hoc and tethering, with benefits and drawbacks				
2.2.2	define speed, bandwidth, throughput, scalability, latency, error rate, packet loss, availability and jitter				
2.2.2	calculate transfer time, throughput, packet loss % and availability %				
2.1.2	describe the purpose and features of Ethernet, Wi-Fi, Bluetooth, cellular, infrared and ZigBee				
2.1.3	name the seven OSI layers in order and describe the role of each				
2.1.3	describe the four TCP/IP layers and map them to OSI				
2.1.3	explain the roles of VOIP and SIP				
2.2.5	compare IPv4 and IPv6; shorten an IPv6 address correctly				
2.2.5	explain static vs dynamic addresses and the role of DHCP				
2.2.5	describe a MAC address and compare it with an IP address				
2.2.4	draw a labelled network design that meets every requirement and takes account of location				
2.3.1	explain the motives for attacks, the main threats, open-network risks and the impact on individuals and organisations				
2.3.2	explain firewalls, security settings, anti-malware, access rights, authentication, encryption and physical controls				

Fact-check summary — where this handbook differs from the textbook

Every topic in the textbook pages supplied was checked. Most of the textbook is accurate. These are the places where it is simplified, out of date or wrong, with what to write in the exam.

Topic (textbook page)	Textbook says	What is correct / what to write
IPv6 shortening (p.57)	"A set of four 0 digits can be replaced by ::"; example `AB10:F1B1::AF00:3121::8998:FF22`	`::` replaces *one or more* consecutive zero groups and may be used **only once**. The example is invalid (RFC 4291).
TCP/IP history (p.54)	Developed by the US DoD in the 1960s	ARPANET began 1969 (DoD-funded); TCP/IP designed by Cerf and Kahn 1973–74; adopted on ARPANET 1 Jan 1983. Still older than OSI (1984).
TCP/IP transport (p.55)	Transport layer uses TCP	Also UDP — used by live streaming, games and VOIP voice data.
Wi-Fi 5 GHz (p.50)	5 GHz developed in 2009	802.11a used 5 GHz from 1999; 2009 is 802.11n (dual-band).
Fibre speed (p.39)	Up to 1 Gbps	Typical home package; fibre links routinely carry 10–400 Gbps. Say "highest bandwidth of wired media".
GPS (p.41)	Triangulation with at least three satellites	Trilateration; normally at least four satellites (fourth corrects the clock). GPS receivers only receive.
Modem (p.34)	Built into all modern internet devices	Phones have cellular modems; most PCs/laptops use the modem in the home router.
Jitter (p.48)	"Disruption" when data is transmitted	Variation in latency / in packet arrival times (Pearson mark scheme wording).
NEC infrared (p.53)	"national electrical code (NEC)"	NEC is the company NEC Corporation; call it the NEC infrared protocol.
ZigBee and Alexa (p.51)	Voice-controlled lights mean ZigBee was used	Only if the bulbs are ZigBee; many use Wi-Fi or Bluetooth.

Sources: Pearson Edexcel IAL Information Technology specification (2018) and WIT11 June 2024 mark scheme; IETF RFC 4291 and RFC 5952 (IPv6 addressing); IEEE 802.11 and 802.3 standards; IEEE Milestone "TCP Enables the Internet, 1974"; Bluetooth SIG; Connectivity Standards Alliance (ZigBee); gps.gov.

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Glossary

Term	Meaning
Access rights	Restrictions placed on who can see and edit data.
Ad hoc network	A temporary wireless network where devices connect directly, without a router or WAP.
Anti-malware	Software that scans a device and removes or quarantines malware.
Availability	The percentage of time a network is available for use, out of the time it should have been.
Bandwidth	The maximum amount of data that can be transferred in a set amount of time.
Biometrics	Using physical features (face, fingerprint, voice) to authenticate a user.
Bluetooth	Short-range (≈10 m) wireless protocol using radio waves at 2.4 GHz between devices close together.
Bridge	Hardware that connects one LAN to another LAN to form one larger LAN.
Client	A computer that is not a point of control; accesses data and services on a server.
Client–server network	A network with central servers as a point of control and clients that connect to them.
Data collision	When two devices send data at the same time and the packets collide.
Data packet	The unit of data containing the information to be transmitted across a network.
DHCP	Protocol that automatically assigns (leases) IP addresses to devices.
Encryption	Scrambling data into ciphertext with a key so it cannot be understood if intercepted.
Error rate	The number of errors in a network, usually as a percentage.
Ethernet	A protocol used to connect devices in a LAN using a wired connection.
Fibre optic cable	A cable made of fine strands of glass or plastic that carries data as light.
Firewall	Hardware or software that checks network traffic against rules and blocks what is not allowed.
Gateway	Hardware that translates between two networks using different protocols.
Ground station	A base on Earth that sends and receives radio waves to and from a satellite.
Hub	Hardware that connects devices in a LAN and sends every packet to all of them.
IP address	A unique address given to a device when it connects to a network; used by routers.
Jitter	The variation in latency / packet arrival times.
LAN	A network within a single building or site.
Latency	The delay in transmission of data across a network.
MAC address	A unique address given to a NIC at manufacture; used by switches.
Malware	Malicious software designed to damage a device or its data.
Modem	Converts between analogue and digital signals.
NIC / WNIC	Hardware built into a device that allows a wired / wireless network connection.

OSI model	A seven-layer theoretical framework that is the basis of many protocols.
Packet loss	The percentage of packets sent that did not arrive.
Pairing	The connection made between two devices using Bluetooth.
Peer / peer-to-peer network	A network with no server; every computer (peer) is equal.
Phishing	Fake messages or websites that trick users into revealing personal details.
Powerline	Adapters that plug into power sockets and send data over the electrical wiring.
Protocol	A set of rules that govern the transfer of data.
Repeater	Regenerates a weak signal so it can travel further.
Router	Reads IP addresses to send data to the correct network or device.
Routing table	A database of known routes across a network.
Satellite	A device orbiting the Earth that uses radio waves to transmit data.
Scalability	How well a network's capacity can be increased or decreased to meet demand.
Security settings	Ways a user controls who can see or access data stored in an account.
SIP	Session Initiation Protocol — starts, manages and ends VOIP sessions.
Speed	The rate at which data is transferred between devices.
Switch	Reads MAC addresses to send data only to the correct device.
TCP/IP	The four-layer protocol suite used to transmit data across the internet.
Tethering	Sharing a mobile phone's internet connection with another device.
Throughput	The amount of data actually transferred in a set amount of time.
Two-factor authentication	Logging in with two different types of factor, e.g. password + phone code.
VOIP	Voice over Internet Protocol — sends voice calls over the internet.
WAN	A network spread over several buildings or sites; the internet is the largest.
WAP	Wireless access point — lets wireless devices connect to a wired network.
Wi-Fi	Wireless protocol using radio waves at 2.4 GHz and 5 GHz (and 6 GHz).
ZigBee	Low-power wireless protocol (2.4 GHz, 250 kbps) connecting IoT devices through a hub.

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Quick reference — one-page summary

Components → what they read

Hub	nothing — floods all ports
Switch / bridge	MAC address
Router	IP address + routing table
Gateway	translates protocols
Modem	analogue ↔ digital
Repeater	boosts signal
WAP	wireless ↔ wired

OSI (7 → 1): All People Seem To Need Data Processing

7 Application	HTTP, FTP; user data
6 Presentation	format, encrypt, compress
5 Session	open / maintain / close
4 Transport	segments, errors, re-send
3 Network	packets, routing, IP
2 Data link	frames, MAC (LLC + MAC)
1 Physical	bits, cables, radio

TCP/IP

Application (= OSI 7+6+5) · Transport (TCP) · Internet (IP) · Link (= OSI 2+1)

Network models

Client–server: central security/backup/updates; costly, server = single point of failure. **P2P**: cheap, all equal; less secure, own backups. **Ad hoc**: temporary direct wireless, no router. **Tethering**: share phone's cellular data; battery + data drain.

Communication methods

Ethernet	wired LAN; cheap, secure, reliable
Wi-Fi	2.4 GHz (range) / 5 GHz (speed)
Bluetooth	2.4 GHz, ≈10 m, 7 devices, ≈3 Mbps, pairing
ZigBee	2.4 GHz, 250 kbps, ≈100 m, hub, IoT
Cellular	cells + masts; 2G 0.1 · 3G 42 · 4G 150 Mbps · 5G 10 Gbps
Infrared	line of sight, few m, NEC protocol

Metrics

Bandwidth max capacity · **Throughput** actual · **Speed** rate · **Latency** delay · **Jitter** variation in delay · **Error rate** % errors · **Packet loss** % not received · **Availability** % uptime · **Scalability** grow/shrink capacity.
time = (MB × 8) ÷ Mbps

Addresses

IPv4 32-bit, 4 denary 0–255, dots. **IPv6** 128-bit, 8 hex groups, colons; drop leading zeros; `::` once only. **MAC** 48-bit, 12 hex, set at manufacture. **Static** fixed (servers); **dynamic** via DHCP (DORA).

Security

Motives: fraud, identity theft, IP theft, revenge, protest, fun. **Threats:** hacking, malware (virus, worm, trojan, spyware, ransomware), phishing, DoS, interception, social engineering. **Solutions:** firewall · security settings · anti-malware · access rights/profiles · authentication (password, 2FA, biometrics) · encryption · physical controls.

NOTES

My notes

Use this page for anything your teacher adds in class, questions to ask, or past-paper mistakes to remember.